Genome-scale context, stoichiometric refinement, and neuro-symbolic reasoning for proteome function annotation

Tawfiq Al-Saeed[1], Maxat Kulmanov[1], and Robert Hoehndorf[1,*]

[1]Computer, Electrical and Mathematical Sciences & Engineering Division (CEMSE),
Computational Bioscience Research Center (CBRC), King Abdullah University of Science and Technology (KAUST),
Thuwal 23955-6900, Kingdom of Saudi Arabia

[*]Correspondence: robert.hoehndorf@kaust.edu.sa

Abstract

Protein function predictors score each sequence or structure in isolation, although long-read assemblies now routinely deliver complete chromosomes whose gene order, complex stoichiometry, and metabolic connectivity constrain which functions a proteome can contain. We present Genome-Scale Protein Annotation (GSPA), which treats function assignment as inference over a complete genome. GSPA calibrates and pools per-protein predictors into a proteome-wide probability matrix, refines this matrix with priors over co-directional operons, essential functions, pathways, obligate heteromeric complexes, and taxonomic and anatomical ontology axioms, repairs the metabolic network with positive-cost parsimonious flux balance analysis that attributes reactions competitively to gene loci, and invokes an ontology-guarded language model only on the small fraction of loci that remain ambiguous. We evaluated GSPA on six complete reference genomes withheld from training under prospective time-based and sequence-identity splits against retrained sequence, domain, graph, and protein language model baselines, and in leave-subunit-out, leave-reaction-out, fitness-validated enzyme, and BacDive phenotype benchmarks. Whole-genome inference raised information-accretion-weighted F-measures over the strongest per-protein integrator on every split, removed all taxonomic and anatomical contradictions, located missing complex subunits from operon context, and attributed flux-carrying reactions to divergent enzymes whose physiological roles had been established by transposon mutagenesis. Organismal constraints therefore complement per-protein prediction, and their contribution is largest where sequence similarity to characterized proteins has been removed. GSPA is available at https://github.com/bio-ontology-research-group/gspa.

1 Introduction

Automated protein function annotation assigns biological roles to the protein-coding genes of an organism and underlies comparative genomics, metabolic modeling, and systems biology [1, 2]. We consider the genome-scale version of this problem: the input is the complete set of protein-coding genes $\mathcal{P} = \{p_1, \ldots, p_N\}$ of an organism together with their chromosomal coordinates and strand organization $\mathcal{K}$, the organism's taxonomic lineage τ_{tax}, and reference biochemical and ontological knowledge bases; the output is a joint assignment of Gene Ontology (GO) terms [3, 4] and Enzyme Commission (EC) numbers across $\mathcal{P}$. Because the proteins encoded by a genome act together in one cell, a valid joint assignment satisfies invariants that span many proteins at once: the core functions required in the organism's domain of life are present, metabolic pathways form connected routes that support biomass synthesis [5, 6], obligate heteromeric complexes are annotated with their complete subunit sets [7, 8], and every assigned term is compatible with the organism's lineage and with the anatomy of its clade [9, 10]. These invariants tie the function of each protein to the architecture of the whole genome.

Formulating function prediction over complete genomes is timely because chromosome-scale assemblies have become routine. Short-read sequencing of isolates and metagenomes produced fragmented contigs from which open reading frames were annotated as isolated sequences, which severed operons, biosynthetic gene clusters, and chromosome-wide reaction inventories. Long-read sequencing [11, 12] and repeat-graph assemblers [13] now recover complete or near-complete chromosomes for cultivated isolates and for metagenome-assembled genomes of uncultured lineages, so gene order and complete proteome complements are available to function prediction as inputs.

Standard predictors derive a representation of one protein and transfer annotations from reference databases. Sequence alignment transfers annotations from homologs [14–16]; profile hidden Markov models and domain signatures recognize conserved functional motifs [17, 18]; convolutional, transformer, and graph neural networks learn sequence motifs, domain graphs, and the GO label hierarchy [19–22]; protein language models and semantic entailment embeddings place functionally related proteins close together in representation space [23–27]; and structural graph networks transfer function across

remote fold similarity [28–30]. Because these modalities capture complementary properties, multi-source ensembles such as GOLabeler [31] and NetGO [32, 33] achieve strong protein-centric accuracy in the Critical Assessment of Functional Annotation (CAFA) [1, 2, 34]. Scoring each protein in isolation nevertheless leaves genome-scale information unused: co-transcribed operon neighbors that share a complex or pathway, previously characterized genes on the same chromosome that can anchor newly studied neighbors, proteome-wide stoichiometric and complex completeness requirements, and clade-specific constraints that could stop a false leaf prediction before it propagates upward through the GO hierarchy.

Comparative genomics, constraint-based metabolic modeling, and formal bio-ontologies each supply part of this organismal context. Conserved gene order, co-directional operons, and genomic language models [35–42] capture local synteny within contiguous gene windows, whereas essential-function profiles and complex stoichiometry act across the whole proteome. Metabolic reconstruction pipelines such as Pathway Tools [43], ModelSEED [44], CarveMe [45], gapseq [46], and GapMind [47] identify the reactions a network needs for steady-state flux, and our earlier method METEOR [48] coupled this gap-filling to neural enzyme prediction through a reaction-indicator mixed-integer linear program (MILP). Two properties of that formulation limit its precision at the gene level: a negative objective cost on candidate reactions rewards the solver for adding reactions that carry no flux, and a uniform reaction-to-gene boost elevates every paralog of a selected reaction equally, without regard to operon neighborhood, subunit role, or competition among candidate loci. In bio-ontologies, GO taxon constraints [9] and the axioms that link GO processes to the Uberon anatomy ontology [10, 49], the Plant Ontology (PO) [50], and the Cell Ontology (CL) [51] state which structures and processes can exist in a clade; applying them before upward propagation, and relaxing a forbidden term to its nearest admissible ancestor, keeps the conserved part of an annotation while removing the cross-clade error. Finally, for loci that remain ambiguous after probabilistic and stoichiometric refinement, tool-using language-model agents [52, 53] can weigh heterogeneous domain, operon, and pathway evidence, provided that a symbolic guardrail verifies what they propose.

Here we present Genome-Scale Protein Annotation (GSPA), a four-stage framework that combines per-protein representation learning with whole-genome probabilistic, stoichiometric, and ontological inference (Figure 1). Stage 0 calibrates and pools sequence homology, domain signatures, convolutional networks, and protein language models into a per-protein prior; GSPA-Seq, a stacked per-protein integrator over the same channels, is the strongest per-protein configuration and serves as the reference point for everything that follows. Stage 1 then updates proteome-wide log-odds iteratively under six organism-level priors, and Stage 2A enforces taxonomic and anatomical admissibility before upward propagation. Stage 1 and Stage 2B carry inference beyond homology: reaction-local genomic context (RLGC) ranks candidate genes for a missing reaction by their chromosomal proximity to genes of neighboring reactions, and Stage 2B couples role-aware aggregation of isozymes and subunits with positive-cost lexicographic pFBA [54] and competitive locus attribution against an explicit unassigned-gene hypothesis. Stage 3 resolves residual ambiguities with a tool-calling language model whose proposals are verified against the ontology. We evaluate GSPA on six complete reference genomes withheld from training under prospective time-based and sequence-identity splits, and in leave-subunit-out, leave-reaction-out, anatomical stress, fitness-validated enzyme, and experimental phenotype benchmarks spanning bacteria, archaea, and eukaryotes (Supplementary Table S1). GSPA, its precomputed ontology bridges, and the benchmark workflows are available at `https://github.com/bio-ontology-research-group/gspa`.

2 Results

2.1 Whole-genome inference on withheld complete genomes

Our central question is whether a predictor that sees the complete genome annotates its proteins better than one that sees each protein alone. CAFA target lists distribute isolated sequences, so we constructed a whole-genome variant of the CAFA protocol: six complete prokaryotic reference genomes (*B. subtilis* 168, *P. aeruginosa* PAO1, *M. tuberculosis* H37Rv, *H. pylori* 26695, *Synechocystis* sp. PCC 6803, and *M. jannaschii* DSM 2661; 20,697 coding sequences) are supplied as whole chromosomes, and their experimentally annotated proteins are scored under two prospective time-based splits (1,060 proteins annotated only after the training cutoff t_0) and four homology splits that additionally purge from training every protein sharing more than 100%, 70%, 50%, or 30% identity with any protein of the six genomes (3,315 target proteins). On every split we retrained fourteen per-protein baselines on that split's training set, including re-implementations of DeepFRI, DeepGraphGO, TALE+, SPROF-GO, DeepGO-SE, and DPFunc that take ESM-2 3B embeddings as input, together with the per-protein integrator GSPA-Seq, and then ran whole-genome inference over all 20,697 coding sequences (Table 1 and Supplementary Table S4). Two caveats bound what the homology splits can show: the InterPro signature libraries and the ESM-2 weights were trained on public data that include these proteomes, so purging training proteins constrains the alignment and retrained components only, and an InterPro logistic regression channel is accordingly the most homology-insensitive baseline in the table.

Whole-genome inference improved on the strongest per-protein configuration on all six splits. With pre-t_0 annotations of neighboring genes retained on the chromo-

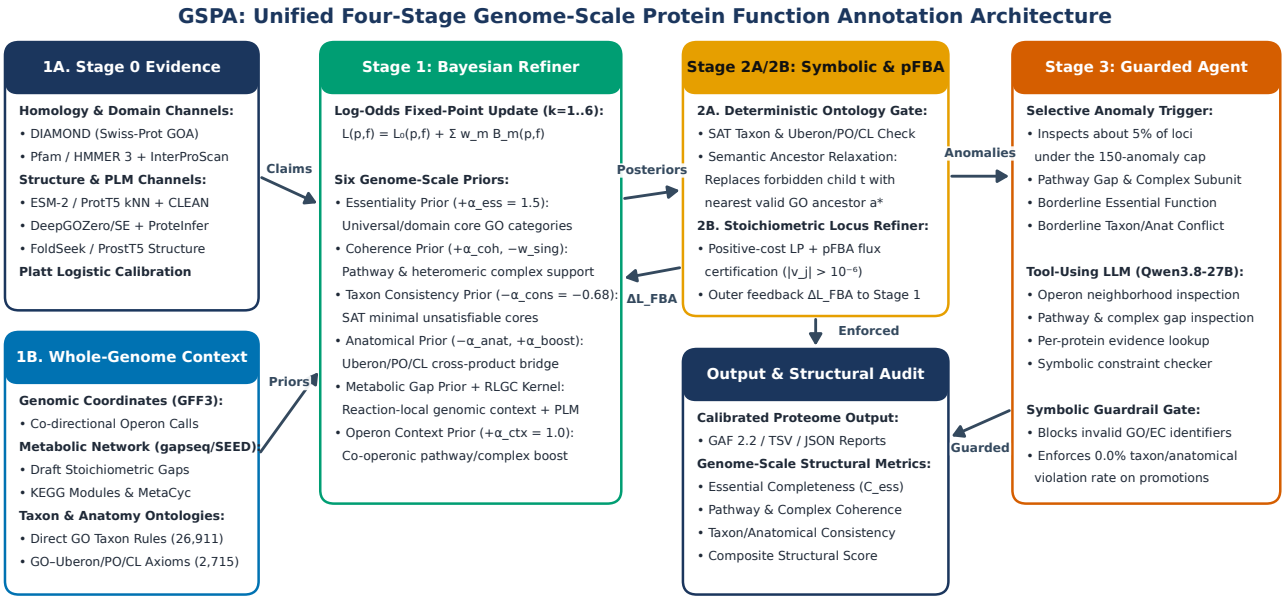


Figure 1. Four-stage architecture of GSPA. Stage 0 calibrates per-channel confidence scores from sequence homology, domain profiles, protein language models, and structural similarity (or from the per-protein stacker GSPA-Seq) and pools them by correlation-grouped Noisy-OR into a proteome-wide probability matrix. Stage 1 refines these probabilities in log-odds space with six whole-proteome priors: core essentiality (B_1), pathway and heteromeric complex coherence (B_2), taxonomic consistency (B_3), cross-ontology anatomical constraints (B_4), metabolic gaps coupled to reaction-local genomic context (B_5), and co-directional operons (B_6). Stage 2A enforces ontology constraints before final ancestor propagation, relaxing forbidden terms to their nearest admissible ancestors and pruning isolated obligate-complex singletons; Stage 2B aggregates isozyme and subunit roles, repairs the metabolic network by positive-cost lexicographic parsimonious flux balance analysis (pFBA), and attributes flux-carrying reactions competitively to candidate loci against an unassigned-gene hypothesis. Stage 3 applies selective tool-mediated language-model reasoning under a deterministic ontology guardrail.

somes (`WG_Time_Rollover`), full GSPA reached a mean information-accretion-weighted f_w of 0.5020 against 0.4949 for GSPA-Seq and 0.4861 for the best retrained external model (SPROF-GO); when every experimentally annotated protein of the six species was removed from training (`WG_Time_Withheld`), the same 1,060 targets were scored at 0.4460, so previously characterized neighbors on the same chromosome contributed +0.056 f_w to the newly characterized genes. The officially released DeepFRI and DeepGO-SE weights are listed on the time-based splits as reference points only; their training data include the withheld proteins, so they are comparable only on the rollover split. As cross-species homologs were purged, DIAMOND transfer collapsed from $f_w = 0.352$ to 0.116 at $\leq 30\%$ identity, the retrained neural baselines fell to 0.26–0.35, and the InterPro channel stayed nearly flat (0.448 → 0.439); Stage 0 pooling stayed between 0.450 and 0.469 across the four homology splits, because channel reliabilities are recalibrated on a validation set for each split and weight shifts from alignment toward the domain and language-model channels as alignment degrades, and full GSPA remained between 0.469 and 0.485. The margin of full GSPA over GSPA-Seq was +0.007 on the rollover split and +0.012 to +0.019 on the homology splits, and the ablation rows locate it: taxonomic and anatomical enforcement before propagation (B_3, B_4 and Stage 2A) raised Cellular Component f_w by 0.015–0.034 while reducing surviving constraint violations to zero, pathway and complex coherence (B_1, B_2) raised Molecular Function f_w by up to 0.017, and operon context (B_6) added smaller gains spread across aspects. The whole-genome gain is therefore modest on well-characterized proteins and largest where sequence similarity has been removed, which is the regime in which per-protein methods have least to offer.

Stage 0 pooling with the three classical channels also compared favorably with production annotation pipelines when all were evaluated under one ancestor-propagated protocol on a panel of 13 complete prokaryotic genomes (11 bacteria and 2 archaea): against experimentally supported annotations, GSPA reached a micro $F_{\max}$ of 0.350, InterProScan 0.328, DIAMOND transfer 0.303, NCBI PGAP [15] 0.212, and metagenomic-deepFRI [55] 0.201, and GSPA exceeded PGAP and DIAMOND on every genome (Figure 2d and Supplementary Table S2; against all GO annotations, 85% of which are electronic and largely InterPro-derived, the same ordering holds at higher values). On a second panel of 21 bacteria for which only alignment inputs were available, a leave-one-genome-out ESM-2 nearest-neighbor transfer across the panel (0.851) exceeded both DIAMOND (0.744) and alignment-only GSPA (0.764), which is why Stage 0 pools embedding channels wherever they can be computed. On enzyme prediction, pooling CLEAN [27], EnzBERT [56], and DeepProZyme [57] and accepting only gap-filling changes supported by all three channels produced net Top-1 gains on 95 bacterial genomes under all four donor-homology

Table 1. Whole-genome withheld CAFA evaluation across prospective time-based splits (`WG_Time_Rollover`, `WG_Time_Withheld`) and sequence-similarity homology splits (`WG_Homology_le100`, `WG_Homology_le70`, `WG_Homology_le50`, `WG_Homology_le30`) across six complete reference proteomes (20,697 **ORFs,** 3,809 **co-directional operons), comparing against official pre-trained and from-scratch retrained external predictors.** Six complete prokaryotic genomes (*B. subtilis* 168, *P. aeruginosa* PAO1, *M. tuberculosis* H37Rv, *H. pylori* 26695, *Synechocystis* sp. PCC 6803, and *M. jannaschii* DSM 2661) are withheld as complete genome inputs (20,697 coding sequences, 3,315 experimentally annotated target proteins, and 1,060 strictly post-t_0 target proteins). Official pre-trained weights of DeepFRI and DeepGO-SE are evaluated on the time-based splits, and all trainable baselines (our re-implementations of DeepFRI, DeepGraphGO, TALE+, SPROF-GO, DeepGO-SE, and DPFunc over ESM-2 3B embeddings, 16-kernel DeepGOCNN and DeepGOPlus, ESM-2 3B k-NN/MLP/C-HMCNN, InterPro-LR, and the out-of-fold GSPA-Seq stacker) are retrained on each split's filtered training corpus (complete 130-row evaluation across all 22 methods in Supplementary Table S4). Columns report official CAFA information-accretion-weighted f_w across Molecular Function (MF), Biological Process (BP), Cellular Component (CC), and aspect mean, unweighted protein-centric $F_{\max}$ aspect mean, and the number of taxon- or anatomy-forbidden protein–term predictions (V) scored at or above the method's f_w-optimal threshold (capped at 0.25). Ablation rows that do not enforce constraints share the GSPA-Seq count by construction. InterPro-LR, the most homology-insensitive baseline, is included on every split.

Whole-Genome Split ($N_{\text{train}} \to N_{\text{eval}}$; 20,697 ORFs)	Pre-trained or Retrained Baseline / Whole-Genome Ablation	f_w (MF)	f_w (BP)	f_w (CC)	Mean f_w	$F_{\max}$ (MF / BP / CC)	Mean $F_{\max}$	Violations (V)
WG_Time_Rollover Prospective $t_0 \to t_1$ w/ Pre-t_0 Chromosomal Context (82,404 → 1,060 Targets)	DeepFRI (Pretrained Official)	0.2655	0.1705	0.2886	0.2415	0.2779/0.2217/0.3642	0.2879	1,300
	DIAMOND BLAST-kNN (CAFA Baseline 2)	0.4829	0.3361	0.3070	0.3753	0.4929/0.3777/0.3487	0.4064	988
	InterPro-LR (Domain Logistic Regression, Retrained)	0.5622	0.4131	0.3680	0.4477	0.5685/0.4586/0.4780	0.5017	916
	DeepFRI (GCN + LM, Retrained)	0.4408	0.2752	0.4366	0.3842	0.4648/0.3535/0.5678	0.4620	61
	DeepGOPlus (16-Kernel CNN + DIAMOND-kNN, Retrained)	0.5499	0.3437	0.3463	0.4133	0.5637/0.3954/0.4452	0.4681	1,026
	DeepGO-SE (ESM-2 3B + ELEmbeddings, Retrained)	0.5068	0.2799	0.4944	0.4270	0.5255/0.3588/0.6108	0.4984	83
	DeepGO-SE (Pretrained Official)	0.5077	0.3014	0.5491	0.4527	0.5264/0.3102/**0.6200**	0.4855	113
	ESM-2 3B C-HMCNN (Hierarchical MCLoss, Retrained)	0.5095	0.3362	0.5141	0.4533	0.5398/0.3975/0.6186	0.5186	237
	TALE+ (Transformer + GO Label Attn + DIAMOND, Retrained)	0.5435	0.3584	0.4624	0.4548	0.5561/0.4236/0.5807	0.5202	216
	DeepGraphGO (InterPro + GraphGCN, Retrained)	0.5051	0.3353	0.5249	0.4551	0.5344/0.3964/0.6180	0.5162	79
	DPFunc (Domain Cross-Attn + GCN, Retrained)	0.5143	0.3476	0.5279	0.4633	0.5364/0.4148/0.6172	0.5228	318
	SPROF-GO (PLM + Hierarchy + Label Diffusion, Retrained)	0.5771	0.3691	0.5122	0.4861	0.5928 /0.4309/0.6163	0.5467	99
	GSPA-Seq (Ours, Per-Protein Integrator)	0.5808	0.3980	0.5059	0.4949	0.5887/0.4528/0.5991	0.5469	938
	GSPA + Operon Genomic Context Only (B_6)	0.5820	0.3980	0.5108	0.4969	0.5888/0.4515/0.5998	0.5467	938
	GSPA + Pathway & Complex Coherence Only (B_1, B_2)	0.5811	0.3977	0.5054	0.4947	0.5889/0.4525/0.5991	0.5468	938
	GSPA + Taxon & Anatomical Constraints Only ($B_3, B_4 + 2A$)	0.5805	0.3981	0.5206	0.4997	0.5885/**0.4529**/0.6043	0.5486	0
	GSPA Full Whole-Genome (Stages 0 + 1 + 2A + 2B)	0.5810	0.3987	0.5262	0.5020	0.5886/0.4517/0.6052	0.5485	0
WG_Time_Withheld Prospective $t_0 \to t_1$ w/ All 6 Genomes Purged from Train (79,880 → 1,060 Targets)	DeepFRI (GCN + LM, Retrained)	0.4257	0.2592	0.2974	0.3274	0.4618/0.3463/0.4568	0.4216	308
	DIAMOND BLAST-kNN (CAFA Baseline 2)	0.4581	0.3022	0.2345	0.3316	0.4699/0.3465/0.2907	0.3690	1,195
	InterPro-LR (Domain Logistic Regression, Retrained)	0.5489	0.4084	0.3383	0.4319	0.5524/0.4542/0.4489	0.4852	1,114
	DeepGO-SE (ESM-2 3B + ELEmbeddings, Retrained)	0.4789	0.2588	0.3560	0.3646	0.5002/0.3392/0.4846	0.4413	257
	DPFunc (Domain Cross-Attn + GCN, Retrained)	0.5034	0.3117	0.3599	0.3917	0.5282/0.3801/0.4917	0.4666	631
	TALE+ (Transformer + GO Label Attn + DIAMOND, Retrained)	0.5411	0.3380	0.3085	0.3959	0.5450/0.4034/0.4748	0.4744	371
	SPROF-GO (PLM + Hierarchy + Label Diffusion, Retrained)	0.5272	0.3378	0.3277	0.3976	0.5442/0.4059/0.4736	0.4746	655
	DeepGraphGO (InterPro + GraphGCN, Retrained)	0.5080	0.3145	0.3716	0.3980	0.5272/0.3838/**0.5149**	0.4753	200
	GSPA-Seq (Ours, Per-Protein Integrator)	0.5687	0.3807	0.3588	0.4361	0.5715/0.4354/0.4717	0.4929	1,426
	GSPA + Pathway & Complex Coherence Only (B_1, B_2)	0.5698	0.3805	0.3586	0.4363	0.5728 /0.4351/0.4717	0.4932	1,426
	GSPA + Taxon & Anatomical Constraints Only ($B_3, B_4 + 2A$)	0.5684	0.3808	0.3924	0.4472	0.5713/**0.4355**/0.4747	0.4938	0
	GSPA Full Whole-Genome (Stages 0 + 1 + 2A + 2B)	0.5657	0.3795	0.3927	0.4460	0.5716/0.4341/0.4747	0.4934	0
WG_Homology_le100 Leave-Whole-Genome-Out: All 6 Genomes Withheld (79,880 → 3,315 Targets)	DeepFRI (GCN + LM, Retrained)	0.4214	0.2951	0.2951	0.3372	0.4690/0.3700/0.4448	0.4279	843
	DIAMOND BLAST-kNN (CAFA Baseline 2)	0.4922	0.3505	0.2139	0.3522	0.5093/0.3884/0.2833	0.3937	4,340
	InterPro-LR (Domain Logistic Regression, Retrained)	0.5865	0.4802	0.2785	0.4484	0.5967/0.5181/0.4067	0.5072	5,168
	DeepGO-SE (ESM-2 3B + ELEmbeddings, Retrained)	0.4926	0.3020	0.3128	0.3691	0.5289/0.3692/0.4595	0.4525	730
	DPFunc (Domain Cross-Attn + GCN, Retrained)	0.5264	0.3460	0.3105	0.3943	0.5622/0.4123/0.4516	0.4754	1,982
	DeepGraphGO (InterPro + GraphGCN, Retrained)	0.5270	0.3508	0.3321	0.4033	0.5620/0.4153/**0.4769**	0.4847	373
	SPROF-GO (PLM + Hierarchy + Label Diffusion, Retrained)	0.5501	0.3878	0.3009	0.4129	0.5806/0.4482/0.4498	0.4929	823
	TALE+ (Transformer + GO Label Attn + DIAMOND, Retrained)	0.5684	0.3865	0.3156	0.4235	0.5849/0.4375/0.4562	0.4929	1,100
	GSPA-Seq (Ours, Per-Protein Integrator)	0.6018	0.4381	0.3248	0.4549	0.6152/0.4872/0.4544	0.5189	4,716
	GSPA + Pathway & Complex Coherence Only (B_1, B_2)	0.6086	0.4386	0.3247	0.4573	0.6205 /0.4869/0.4544	0.5206	4,716
	GSPA + Taxon & Anatomical Constraints Only ($B_3, B_4 + 2A$)	0.6017	0.4384	0.3550	0.4651	0.6151/**0.4875**/0.4569	0.5198	0
	GSPA Full Whole-Genome (Stages 0 + 1 + 2A + 2B)	0.6059	0.4373	0.3627	0.4686	0.6188/0.4860/0.4569	0.5206	0
WG_Homology_le70 All 6 Genomes + Homologs > 70% Identity Purged (79,457 → 3,315 Targets)	DeepFRI (GCN + LM, Retrained)	0.4098	0.2830	0.2686	0.3205	0.4595/0.3567/0.4092	0.4085	2,074
	DIAMOND BLAST-kNN (Filtered ≤ 70%)	0.4842	0.3473	0.2046	0.3454	0.5011/0.3853/0.2771	0.3878	4,355
	InterPro-LR (Domain Logistic Regression, Retrained)	0.5849	0.4816	0.2771	0.4479	0.5939/0.5189/0.4049	0.5059	5,206
	DeepGO-SE (ESM-2 3B + ELEmbeddings, Retrained)	0.4931	0.2998	0.3092	0.3674	0.5298/0.3697/0.4561	0.4519	558
	DPFunc (Domain Cross-Attn + GCN, Retrained)	0.5200	0.3443	0.3086	0.3910	0.5525/0.4139/0.4444	0.4703	1,208
	DeepGraphGO (InterPro + GraphGCN, Retrained)	0.5189	0.3498	0.3381	0.4023	0.5524/0.4114/**0.4843**	0.4827	514
	SPROF-GO (PLM + Hierarchy + Label Diffusion, Retrained)	0.5557	0.3869	0.3294	0.4240	0.5853/0.4465/0.4627	0.4982	1,858
	TALE+ (Transformer + GO Label Attn + DIAMOND, Retrained)	0.5662	0.3867	0.3223	0.4251	0.5857/0.4386/0.4583	0.4942	1,875
	GSPA-Seq (Ours, Per-Protein Integrator)	0.6030	0.4379	0.3389	0.4599	0.6183/**0.4866**/0.4627	0.5225	4,412
	GSPA + Pathway & Complex Coherence Only (B_1, B_2)	0.6132	0.4385	0.3381	0.4633	0.6260 /0.4864/0.4627	0.5250	4,412
	GSPA Full Whole-Genome (Stages 0 + 1 + 2A + 2B)	0.6114	0.4335	0.3769	0.4740	0.6252/0.4831/0.4649	0.5244	0
WG_Homology_le50 All 6 Genomes + Homologs > 50% Identity Purged (76,405 → 3,315 Targets)	DIAMOND BLAST-kNN (Filtered ≤ 50%)	0.4327	0.2930	0.1747	0.3001	0.4534/0.3331/0.2500	0.3455	6,231
	InterPro-LR (Domain Logistic Regression, Retrained)	0.5862	0.4778	0.2764	0.4468	0.5940/0.5146/0.4037	0.5041	4,919
	DeepFRI (GCN + LM, Retrained)	0.4208	0.2792	0.2976	0.3325	0.4701/0.3547/0.4126	0.4125	2,475
	DeepGO-SE (ESM-2 3B + ELEmbeddings, Retrained)	0.4843	0.2882	0.3149	0.3625	0.5254/0.3567/0.4615	0.4479	922
	DeepGraphGO (InterPro + GraphGCN, Retrained)	0.5012	0.3290	0.3169	0.3824	0.5372/0.3935/0.4593	0.4633	446
	DPFunc (Domain Cross-Attn + GCN, Retrained)	0.5041	0.3187	0.3431	0.3886	0.5406/0.3864/**0.4776**	0.4682	2,784
	TALE+ (Transformer + GO Label Attn + DIAMOND, Retrained)	0.5513	0.3523	0.3069	0.4035	0.5744/0.4093/0.4539	0.4792	1,761
	SPROF-GO (PLM + Hierarchy + Label Diffusion, Retrained)	0.5419	0.3567	0.3144	0.4043	0.5746/0.4193/0.4524	0.4821	1,458
	GSPA-Seq (Ours, Per-Protein Integrator)	0.5962	0.4225	0.3411	0.4533	0.6143/0.4729/0.4451	0.5107	4,458
	GSPA + Taxon & Anatomical Constraints Only ($B_3, B_4 + 2A$)	0.5962	0.4234	0.3746	0.4647	0.6143/**0.4732**/0.4474	0.5116	0
	GSPA Full Whole-Genome (Stages 0 + 1 + 2A + 2B)	0.6109	0.4206	0.3860	0.4725	0.6257 /0.4716/0.4474	0.5149	0
WG_Homology_le30 Twilight Zone: All 6 Genomes + Homologs > 30% Purged (61,323 → 3,315 Targets)	DIAMOND BLAST-kNN (Filtered ≤ 30%)	0.1756	0.1026	0.0707	0.1163	0.1872/0.1165/0.1065	0.1367	5,519
	InterPro-LR (Domain Logistic Regression, Retrained)	0.5788	0.4695	0.2681	0.4388	0.5879/0.5040/0.4013	0.4978	3,330
	DeepGOPlus (16-Kernel CNN + DIAMOND-kNN, Retrained)	0.4015	0.1922	0.2344	0.2760	0.4487/0.2452/0.3731	0.3556	3,980
	DeepFRI (GCN + LM, Retrained)	0.3660	0.2348	0.2099	0.2702	0.4280/0.2937/0.3827	0.3681	1,949
	DeepGO-SE (ESM-2 3B + ELEmbeddings, Retrained)	0.4124	0.2475	0.3246	0.3282	0.4652/0.3103/0.4345	0.4033	2,139
	DPFunc (Domain Cross-Attn + GCN, Retrained)	0.4235	0.2665	0.3031	0.3311	0.4729/0.3321/0.4434	0.4162	3,413
	DeepGraphGO (InterPro + GraphGCN, Retrained)	0.4201	0.2771	0.3162	0.3378	0.4690/0.3425/**0.4706**	0.4274	376
	SPROF-GO (PLM + Hierarchy + Label Diffusion, Retrained)	0.4389	0.2625	0.3291	0.3435	0.4868/0.3278/0.4499	0.4215	1,063
	TALE+ (Transformer + GO Label Attn + DIAMOND, Retrained)	0.4549	0.2466	0.3309	0.3442	0.4979/0.3088/0.4527	0.4198	2,787
	ESM-2 3B C-HMCNN (Hierarchical MCLoss, Retrained)	0.4360	0.2534	0.3633	0.3509	0.4822/0.3136/0.4615	0.4191	1,245
	GSPA-Seq (Ours, Per-Protein Integrator)	0.6138	0.4412	0.3636	0.4729	0.6234 /**0.4799**/0.4279	0.5104	3,268
	GSPA Full Whole-Genome (Stages 0 + 1 + 2A + 2B)	0.6067	0.4375	0.4098	0.4847	0.6185/0.4778/0.4337	0.5100	0

masking tiers, whereas accepting unfiltered METEOR changes lowered Top-1 accuracy because reaction-level boosts promote non-cognate paralogs (Supplementary Tables S5 and S6). We next asked which organism-level invariants the whole-genome priors change.

2.2 Whole-genome priors act on organism-level invariants and preserve protein-level accuracy

Each Stage 1 prior targets one invariant of a proteome, so we asked whether each prior changes its own invari-

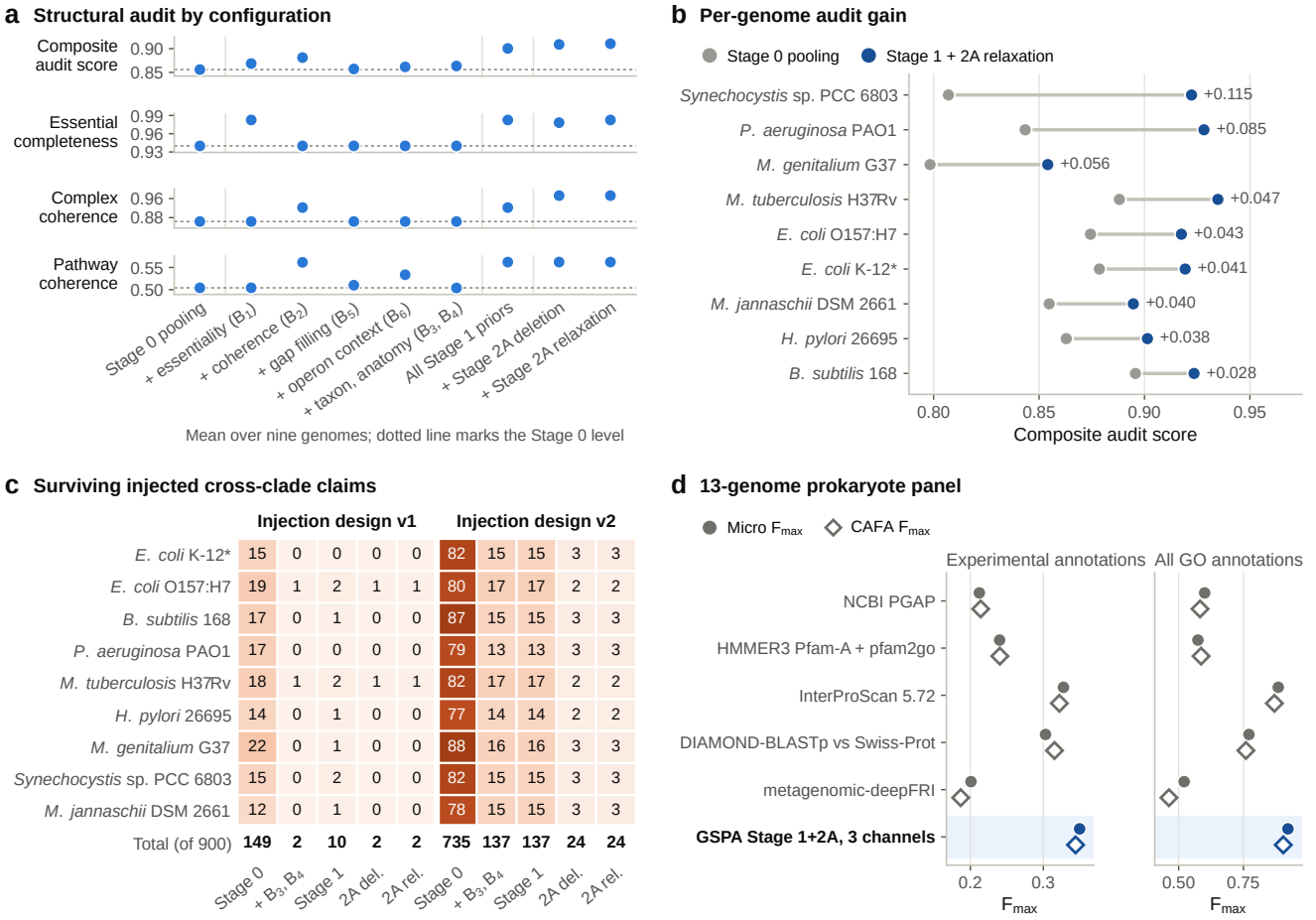


Figure 2. Stage 1 prior and Stage 2A ablation, cross-clade perturbation, and the reference panel. (**a**) Composite structural audit score, essential function completeness, complex coherence, and pathway coherence across the nine ablation configurations, averaged over nine bacterial and archaeal reference genomes; the dotted line marks the Stage 0 level. (**b**) Per-genome composite audit score under Stage 0 pooling and after Stage 1 and Stage 2A relaxation; the asterisk marks the calibration genome, *E. coli* K-12 MG1655. (**c**) Injected cross-clade claims per genome (100 per genome) surviving the 0.50 threshold under both injection designs (Supplementary Table S18) after Stage 0 pooling, the soft taxonomic and anatomical priors, all Stage 1 priors, and Stage 2A deletion or relaxation. (**d**) Micro and protein-centric CAFA $F_{\max}$ of five external annotation pipelines and three-channel GSPA on the 13-genome prokaryote panel against experimental annotations and against all GO annotations.

ant without degrading protein-level accuracy. We ran a nine-configuration ablation on nine bacterial and archaeal reference proteomes, adding priors one at a time to Stage 0 and finally applying Stage 2A either by deleting forbidden terms or by relaxing them to admissible ancestors (Figure 2a,b and Table 2). The priors acted selectively: the essentiality prior raised core functional completeness from 0.940 to 0.983 and left every other column unchanged; the coherence prior raised pathway completeness from 0.504 to 0.562 and heteromeric complex completeness from 0.864 to 0.922; and the taxonomic and anatomical priors removed 16 of the 25 endogenous violations in Stage 1, after which Stage 2A removed the remaining nine. The composite structural audit score [58] rose from 0.856 to 0.911, improving on all nine genomes ($p = 0.0039$, two-sided exact sign test; $0.853 \rightarrow 0.910$ on the eight genomes other than the calibration genome *E. coli* K-12). Protein-centric CAFA $F_{\max}$ moved by less than 0.002 across all configurations, which is expected because roughly 85% of proteins in these reference genomes already carry a high-confidence homolog in Stage 0 and the priors concentrate their updates on the remainder. Relaxation was preferable to deletion in Stage 2A on every column in which the two differed, since it transfers the support of a forbidden term to the ancestors that remain valid instead of discarding it. Applying the same audit to the output of the external pipelines on the 13-genome panel (Table 2, top rows) gives composite scores of 0.72–0.80 with 2–113 taxonomic violations, against 0.89 and none for three-channel GSPA, and when Stage 1 and 2A refinement is applied on top of any of those pipelines' outputs, all but 0–3 of 100 injected cross-clade claims per genome are removed. Because the audit score measures the same invariants that Stages 1 and 2 optimize, we treat it as an internal consistency check and rely on the held-out benchmarks in the remaining sections for accuracy.

Organism-level constraints should also protect a proteome against high-scoring cross-clade transfer, which passes protein-level calibration unchanged. We injected 100 eukaryotic developmental and clade-restricted protein–term claims into each of the nine prokaryotic proteomes,

Table 2. External annotation pipelines on a 13-genome prokaryote panel, and ablation of Stage 1 priors and Stage 2A enforcement across nine bacterial and archaeal reference proteomes. Top rows compare NCBI PGAP, HMMER3/Pfam-A, InterProScan, DIAMOND transfer, metagenomic-deepFRI, and three-channel GSPA on 13 complete prokaryotic genomes under one ancestor-propagated evaluator against all GO annotations (experimental-only annotations in parentheses); their structural audit is computed at threshold 0.50 on each pipeline's own output, and the perturbation column reports the mean number of 100 injected cross-clade claims per genome that survive when Stage 1 and 2A refinement is applied on top of that pipeline's output (all 100 survive the threshold alone). Configurations A1–A9 add priors cumulatively to Stage 0 pooling on *E. coli* K-12 (calibration genome), *E. coli* O157:H7, *B. subtilis* 168, *P. aeruginosa* PAO1, *M. tuberculosis* H37Rv, *H. pylori* 26695, *M. genitalium* G37, *Synechocystis* sp. PCC 6803, and *M. jannaschii* DSM 2661; the last row gives the mean over the eight non-calibration genomes. Protein-centric CAFA $F_{\max}$ and pair-level micro $F_{\max}$ are evaluated against all GO annotations (experimental-only annotations in parentheses), alongside experimental information-content recall (coverage in parentheses). Consistency is the mean over genomes of $\max(0, 1 - 0.01V_g)$ for V_g violations in genome g (total V in parentheses). The last column gives the composite structural audit score and the mean number of injected cross-clade claims per genome surviving the 0.50 threshold out of 100 (suppression rate over 900 claims in parentheses); perturbation runs were performed for A1 and A6–A9, and A2–A5 inherit the A1 value because those priors do not act on admissibility.

Baseline Pipeline or Ablation Configuration	CAFA $F_{\max}$ (All / Exp)	Micro $F_{\max}$ (All / Exp)	Exp. IC Rec. (Cov.)	Essential C_{ess}	Pathway Coh.	Complex Coh.	Consistency (V)	Audit Score / Perturb. Surv. (Supp.)
External Annotation Pipelines & GSPA on the 13-Genome Prokaryote Panel (11 Bacteria, 2 Archaea; One Evaluator, GSPA A9 Applied on Top for Perturbation)								
NCBI PGAP (13-Genome Prokaryote Panel)	0.5825 (0.2141)	0.6003 (0.2123)	0.1701 (0.4799)	0.8260	0.3694	0.7821	0.9900 (13)	0.7755 / 0.00 (100.0%)
HMMER3 Pfam-A + pfam2go (13-Genome Panel)	0.5865 (0.2404)	0.5743 (0.2401)	0.1498 (0.4892)	0.7632	0.2501	1.0000	0.9985 (2)	0.7636 / 3.00 (97.0%)
InterProScan 5.72 (13-Genome Panel)	0.8694 (0.3220)	0.8843 (0.3276)	0.3380 (0.7245)	0.8929	0.3671	0.7179	0.9877 (16)	0.7968 / 3.00 (97.0%)
DIAMOND-BLASTp vs Swiss-Prot (13-Genome Panel)	0.7595 (0.3153)	0.7712 (0.3030)	0.3928 (0.4954)	0.8994	0.4197	0.7165	0.9131 (113)	0.7888 / 2.85 (97.2%)
metagenomic-deepFRI (13-Genome Panel)	0.4626 (0.1867)	0.5211 (0.2008)	0.0867 (0.3498)	0.6146	0.1811	1.0000	0.9938 (8)	0.7167 / 1.00 (99.0%)
GSPA Stage 1+2A, 3 Channels (13-Genome Panel)	0.9037 (0.3442)	0.9216 (0.3498)	0.4978 (0.8019)	0.9681	0.4987	1.0000	1.0000 (0)	0.8944 / 2.92 (97.1%)
9-Genome Stage 1 Prior & Stage 2A Enforcement Ablation ($N = 9$ Bacterial & Archaeal Reference Genomes)								
A1: Stage 0 Noisy-OR Baseline	0.8041 (0.2361)	0.7672 (0.2836)	0.3661 (0.7973)	0.9400	0.5041	0.8636	0.9722 (25)	0.8559 / 81.7 (18.3%)
A2: + Essentiality Prior (B_1)	0.8040 (0.2361)	0.7683 (0.2838)	0.3661 (0.7973)	0.9826	0.5041	0.8636	0.9722 (25)	0.8686 / as A1
A3: + Coherence Prior (B_2)	0.8028 (0.2357)	0.7671 (0.2815)	0.3705 (0.8003)	0.9400	0.5616	0.9222	0.9722 (25)	0.8812 / as A1
A4: + Gap-Filling Prior (B_3)	0.8039 (0.2361)	0.7685 (0.2838)	0.3664 (0.7975)	0.9400	0.5100	0.8636	0.9722 (25)	0.8573 / as A1
A5: + Operon Context Prior (B_6)	0.8037 (0.2356)	0.7675 (0.2815)	0.3682 (0.7973)	0.9400	0.5336	0.8636	0.9722 (25)	0.8616 / as A1
A6: + Consistency & Anatomical (B_3, B_4)	0.8060 (0.2371)	0.7711 (0.2851)	0.3655 (0.7968)	0.9400	0.5038	0.8636	0.9900 (9)	0.8635 / 15.2 (84.8%)
A7: All Stage 1 Priors (B_1–B_6)	0.8043 (0.2368)	0.7703 (0.2828)	0.3708 (0.8000)	0.9826	0.5621	0.9222	0.9900 (9)	0.9005 / 15.2 (84.8%)
A8: Stage 1 + Stage 2A Removal	0.8053 (0.2378)	0.7710 (0.2852)	0.3708 (0.8000)	0.9782	0.5622	0.9722	1.0000 (0)	0.9090 / 2.7 (97.3%)
A9: Stage 1 + Stage 2A Relaxation	0.8056 (0.2382)	0.7733 (0.2863)	0.3747 (0.8057)	0.9826	0.5622	0.9722	1.0000 (0)	0.9106 / 2.7 (97.3%)
Out-of-Sample 8-Genome Mean (A1 → A9)	0.8143 → **0.8160**	0.7764 → **0.7830**	0.3565 → **0.3658**	0.9325 → **0.9805**	0.4904 → **0.5514**	0.8743 → **0.9844**	0.9738 → **1.0000**	0.8531 → **0.9096**

drawing the terms from the same taxon and anatomy constraint tables that the priors consult, under two injection designs (Figure 2c and Supplementary Table S18). With a single injected claim at a raw score in $[0.55, 0.85]$, unconstrained pooling let 16.6 injected claims per genome survive the 0.50 threshold and the Stage 1 soft priors alone removed all but 0.2, so Stage 2A had nothing left to do. With a harder design, raw scores in $[0.62, 0.90]$ and a second corroborating claim on 60% of the pairs, 81.7 claims per genome survived unconstrained pooling, the soft priors reduced the survivors to 15.2, and Stage 2A to 2.7, while micro $F_{\max}$ on the perturbed proteomes rose from 0.763 to 0.772. The soft priors therefore suffice against isolated cross-clade claims, and deterministic enforcement is what removes corroborated ones. Because the injected terms come from the constraint tables, this experiment establishes that enforcement fires on known cross-clade terms at these scores; how often such transfers arise in practice is measured on unperturbed eukaryotic proteomes in Section 2.4.

Applying the same structural audit to six external GO predictors across 1,000 RefSeq bacterial genomes shows how differently per-protein methods respect organism-level invariants even when their essential-function coverage is similar (87%–100% of 22 core categories): MetaCyc pathway completeness ranges from 0.2% to 40% and taxon-violating protein–term rows from about two per genome for InterProScan to about 40,000 for DeepFRI (Supplementary Table S8). GSPA run on the same 1,000 genomes shows what the priors change at this scale: Stage 0 pooling of DIAMOND, Pfam, and InterProScan claims reaches 44% MetaCyc completeness with 43 taxon-violating rows per genome, and Stage 1 with Stage 2A relaxation raises completeness to 65% and leaves 2.2 such rows, all of them terms that the audit's constraint table forbids and the table bundled with GSPA does not. On 88 proteomes spanning Bacteria, Archaea, Eukaryota, and Viruses, whole-genome refinement produced small positive mean gains in $F_{\max}$ whose 95% paired bootstrap intervals excluded zero for Bacteria and Viruses and included it for the smaller Archaea and Eukaryota cohorts (Supplementary Table S9). Genome-wide complex coherence, however, only asks whether partner subunits exist somewhere in the proteome; we next asked whether chromosomal architecture can identify which gene encodes a missing subunit.

2.3 Operon context localizes missing subunits of heteromeric complexes

When one subunit of an obligate complex has diverged into the twilight zone, several proteins may carry a weak match to the missing role, and the question is whether complex stoichiometry together with operon organization can single out the right one. We constructed 175 leave-subunit-out trials in six prokaryotic genomes: the true subunit's input score was set to 0.50 (two-subunit complexes) or 0.45, its partner subunits were left at high confidence, and five decoy proteins received synthetic domain-channel claims of 0.42–0.49 for the same role, drawn either from unrelated loci or, in a second design, from the true subunit's own operon or co-directional run (Figure 3a, Table 3, and Supplementary Table S19). Without priors, the attenuated subunit ranked first in 56% of trials, and after Stage 0 pooling no attenuated subunit remained above the 0.50 threshold; the real DIAMOND channel, capped at the attenuation level, ranked it first in 85% because decoys rarely carry a real claim for the complex term, which is a limit of synthetic decoys that both designs share. With decoys from unrelated loci, the genome-wide coherence prior restored the subunit above threshold in half of the

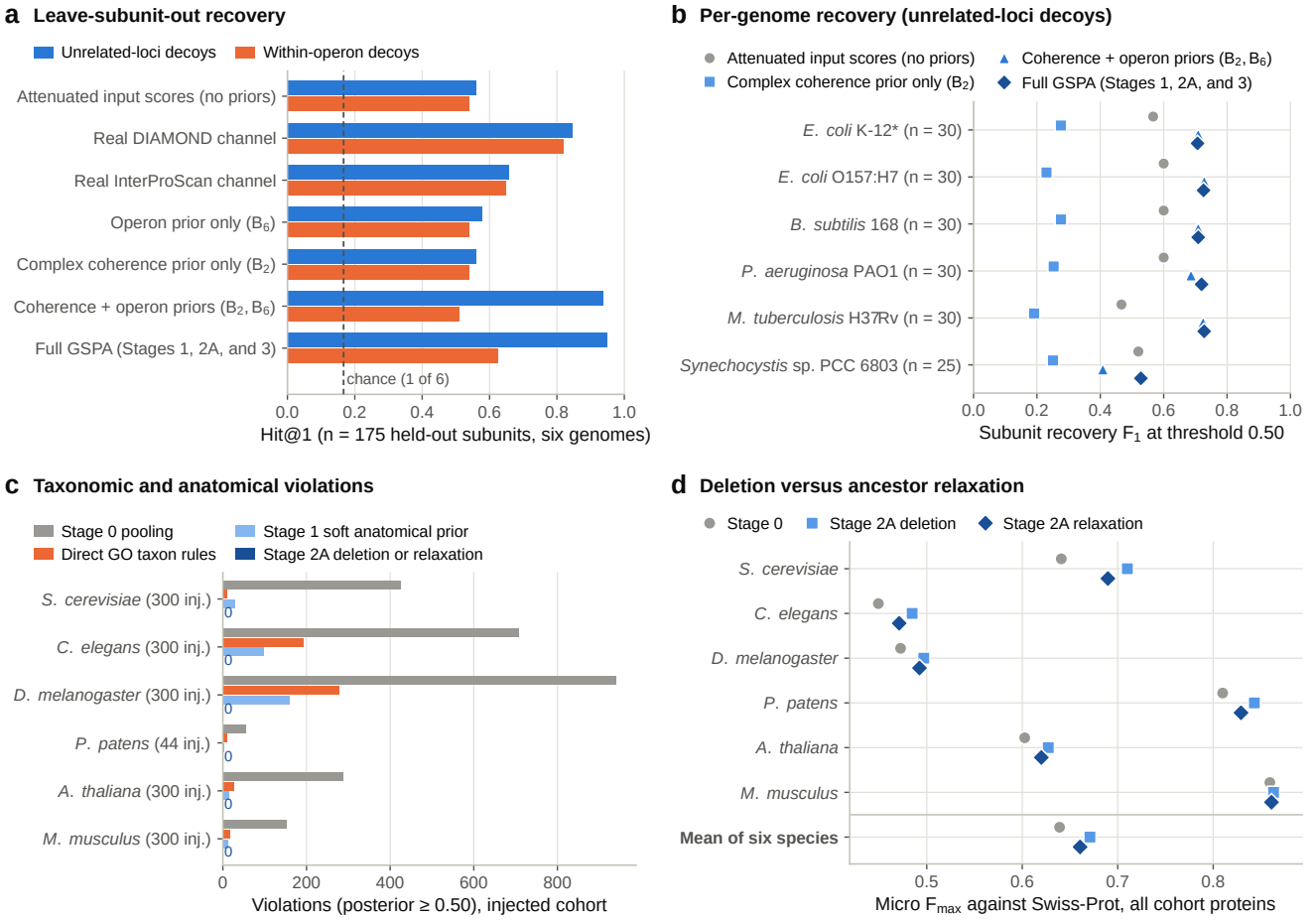


Figure 3. Heteromeric complex completion and eukaryotic anatomical constraint enforcement. (**a**) Leave-subunit-out Hit@1 across 175 held-out subunits in six prokaryotic genomes for each configuration of Table 3, with decoys drawn from unrelated loci or from the true subunit's own operon (chance = 0.167). (**b**) Per-genome subunit recovery F_1 at threshold 0.50 with unrelated-locus decoys; the asterisk marks the calibration genome (*E. coli* K-12). (**c**) Violations remaining in the injected cohort of six eukaryotes, built on real DIAMOND and Pfam claims, under Stage 0 pooling, direct GO taxon rules, the Stage 1 soft anatomical prior, and Stage 2A enforcement. (**d**) Micro $F_{\max}$ against Swiss-Prot annotations over all cohort proteins under Stage 0 pooling, Stage 2A deletion, and Stage 2A relaxation.

trials but left Hit@1 unchanged, since every candidate with a matching role receives the same genome-wide boost, and the operon prior alone moved Hit@1 only to 0.58; the two together raised Hit@1 to 0.94 ($b = 69$, $c = 3$, $p = 2.6 \times 10^{-17}$, exact McNemar test), subunit F_1 at 0.50 to 0.67, and complex coherence from 0.39 to 0.72, and the guarded Stage 3 agent added little (0.95). With decoys from the same operon, however, neither prior separates the true subunit from its neighbors (Hit@1 0.51 against 0.54 for the raw inputs), and only Stage 3, which reads domain architecture and pathway membership, recovered part of the difference (0.63; $b = 32$, $c = 12$, $p = 0.004$). Complex stoichiometry identifies which assembly is incomplete and operon organization identifies which locus completes it when the competitors lie outside that locus; distinguishing paralogs within one operon requires evidence beyond gene order.

Lineages whose complexes are dispersed across the chromosome test this mechanism at its limit. In the cyanobacterium *Synechocystis* sp. PCC 6803, where photosystem, phycobilisome, and NAD(P)H dehydrogenase subunits are scattered among separate loci, the combined priors raised Hit@1 only from 0.52 to 0.64 over 25 trials with unrelated-locus decoys, against 0.97–1.00 in the five species with compact operons (Figure 3b), and Stage 3 raised it to 0.72. Operon synteny resolves contiguous assemblies; dispersed assemblies and within-operon paralogs need evidence beyond local gene order. Multicellular eukaryotes raise a different organism-level invariant, that annotations must respect the body plan of the clade, which we examined next.

2.4 Anatomical constraints correct eukaryotic annotations and relaxation keeps valid ancestors

We first tested whether real predictions on ordinary eukaryotic proteomes contain anatomical contradictions at all. For six eukaryotes (*Saccharomyces cerevisiae*, *Caenorhabditis elegans*, *Drosophila melanogaster*, *Physcomitrium patens*, *Arabidopsis thaliana*, and *Mus musculus*) we took the DIAMOND and Pfam claims produced for 6,155 Swiss-

Table 3. Leave-subunit-out recovery of 175 obligate heteromeric complex subunits in six prokaryotic genomes under two decoy designs. In each trial the raw score of the true subunit's claim for the complex term is set to 0.50 (two-subunit complexes) or 0.45 (larger complexes), its partner subunits are raised to raw scores ≥ 0.85, and five decoy proteins without annotation to the complex term receive synthetic domain-channel claims drawn from $U(0.42, 0.49)$. In the *unrelated-loci* design the decoys are random proteins from loci unrelated to the complex; in the *within-operon* design the same decoy scores are given to proteins from the true subunit's own operon, completed from its co-directional gene run and, where still fewer than five candidates were available (71 of 175 trials), from the nearest genes on the same contig. Real-channel rows rank the candidates by their real DIAMOND or InterProScan/Pfam claims for the complex term, capped at the trial's attenuation level. Hit@1 scores ties at the top as 1/ties; precision, recall, and F_1 are computed at threshold 0.50 among the six candidates, and complex coherence is genome-wide. Stage 3 is the guarded tool-calling agent (Qwen3.8-27B served by vLLM, temperature 0, single run with seed 42). Stage 2A alone changes no score on this benchmark (its output is identical to coherence + operon priors) and is not shown. Rows are given for all six species ($n = 175$), the five non-calibration species ($n = 145$), and *Synechocystis* sp. PCC 6803 alone ($n = 25$); bold marks the column maximum within each design and cohort. Exact McNemar tests on the strict Hit@1 indicator are given in the Supplementary Information.

Cohort	Baseline or Method Configuration	Subunit Trials (n)	Hit@1	Precision@0.5	Recall@0.5	F_1@0.5	Complex Coherence
Decoy design: unrelated loci							
All 6 species	Attenuated input scores (no priors)	175	0.5600	0.5600	0.5600	0.5600	0.5443
	Real DIAMOND channel (capped at attenuation)	175	0.8476	0.3886	0.3886	0.3886	0.4742
	Real InterProScan channel (capped at attenuation)	175	0.6571	0.3143	0.3143	0.3143	0.4812
	Operon prior only (B_6)	175	0.5771	0.0171	0.0171	0.0171	0.3901
	Complex coherence prior only (B_2)	175	0.5600	0.1729	0.5029	0.2459	0.7107
	Coherence + operon priors (B_2, B_6)	175	0.9371	0.5951	0.9257	0.6680	0.7170
	Full GSPA (Stages 1, 2A, and 3)	175	0.9486	0.6196	0.9429	0.6906	0.7170
Out-of-sample (5 species)	Attenuated input scores (no priors)	145	0.5586	0.5586	0.5586	0.5586	0.5265
	Real DIAMOND channel (capped at attenuation)	145	0.8851	0.3862	0.3862	0.3862	0.4682
	Real InterProScan channel (capped at attenuation)	145	0.6839	0.3241	0.3241	0.3241	0.4661
	Operon prior only (B_6)	145	0.5793	0.0207	0.0207	0.0207	0.3740
	Complex coherence prior only (B_2)	145	0.5586	0.1686	0.4897	0.2397	0.6941
	Coherence + operon priors (B_2, B_6)	145	0.9310	0.5886	0.9103	0.6595	0.7016
	Full GSPA (Stages 1, 2A, and 3)	145	0.9448	0.6185	0.9310	0.6873	0.7016
Synechocystis PCC 6803	Attenuated input scores (no priors)	25	0.5200	0.5200	0.5200	0.5200	0.4386
	Real DIAMOND channel (capped at attenuation)	25	0.8667	0.2000	0.2000	0.2000	0.3846
	Real InterProScan channel (capped at attenuation)	25	0.6667	0.2000	0.2000	0.2000	0.4308
	Operon prior only (B_6)	25	0.5200	0.0000	0.0000	0.0000	0.3587
	Complex coherence prior only (B_2)	25	0.5200	0.1813	0.4800	0.2507	0.6125
	Coherence + operon priors (B_2, B_6)	25	0.6400	0.3393	0.6400	0.4080	0.6125
	Full GSPA (Stages 1, 2A, and 3)	25	0.7200	0.4593	0.7600	0.5280	0.6125
Decoy design: within operon							
All 6 species	Attenuated input scores (no priors)	175	0.5400	0.5467	0.5600	0.5505	0.5443
	Real DIAMOND channel (capped at attenuation)	175	0.8190	0.3752	0.3886	0.3790	0.4742
	Real InterProScan channel (capped at attenuation)	175	0.6486	0.3114	0.3143	0.3124	0.4812
	Operon prior only (B_6)	175	0.5400	0.0045	0.0171	0.0070	0.3901
	Complex coherence prior only (B_2)	175	0.5400	0.1729	0.5029	0.2459	0.7107
	Coherence + operon priors (B_2, B_6)	175	0.5114	0.2109	0.9257	0.3305	0.7170
	Full GSPA (Stages 1, 2A, and 3)	175	0.6257	0.2342	0.9486	0.3540	0.7170
Out-of-sample (5 species)	Attenuated input scores (no priors)	145	0.5345	0.5425	0.5586	0.5471	0.5265
	Real DIAMOND channel (capped at attenuation)	145	0.8540	0.3701	0.3862	0.3747	0.4682
	Real InterProScan channel (capped at attenuation)	145	0.6770	0.3207	0.3241	0.3218	0.4661
	Operon prior only (B_6)	145	0.5345	0.0054	0.0207	0.0085	0.3740
	Complex coherence prior only (B_2)	145	0.5345	0.1686	0.4897	0.2397	0.6941
	Coherence + operon priors (B_2, B_6)	145	0.5000	0.2079	0.9103	0.3247	0.7016
	Full GSPA (Stages 1, 2A, and 3)	145	0.5690	0.2357	0.9379	0.3526	0.7016
Synechocystis PCC 6803	Attenuated input scores (no priors)	25	0.5200	0.5200	0.5200	0.5200	0.4386
	Real DIAMOND channel (capped at attenuation)	25	0.8467	0.2000	0.2000	0.2000	0.3846
	Real InterProScan channel (capped at attenuation)	25	0.6467	0.2000	0.2000	0.2000	0.4308
	Operon prior only (B_6)	25	0.5200	0.0000	0.0000	0.0000	0.3587
	Complex coherence prior only (B_2)	25	0.5200	0.1813	0.4800	0.2507	0.6125
	Coherence + operon priors (B_2, B_6)	25	0.4800	0.1893	0.6400	0.2720	0.6125
	Full GSPA (Stages 1, 2A, and 3)	25	0.4800	0.3093	0.7600	0.3920	0.6125

Table 4. Taxonomic and anatomical constraint enforcement and ancestor relaxation across six eukaryotes. The stress cohort of each species contains every protein that carries at least one real blind predictor claim from the multikingdom benchmark (DIAMOND transfer from a reference set without same-species or same-genus hits, and Pfam-to-GO mapping). One cross-clade developmental claim (raw score 0.82–0.95) is injected into 300 randomly chosen cohort proteins per species (44 for *P. patens*), and 20% of the true, species-appropriate developmental claims scoring above 0.62 are attenuated into $[0.55, 0.62]$. Violation columns give the protein–term predictions with posterior ≥ 0.50 whose term is forbidden for the species under unconstrained Stage 0 pooling, direct GO taxon rules, the Stage 1 soft anatomical prior, and Stage 2A enforcement by deletion and by ancestor relaxation (percentage of Stage 0 violations eliminated in parentheses). Pooled micro $F_{\max}$ is evaluated against the unmodified Swiss-Prot annotations on all cohort proteins (propagated recall in parentheses) and on the injected proteins, under Stage 0 pooling, Stage 2A deletion, and Stage 2A relaxation. IC retained is the mean information content of the terms above 0.50 after relaxation relative to Stage 0 on the injected proteins. Developmental rescue counts attenuated claims recovered above 0.50 under relaxation. The last row runs the unperturbed real claims through the same configurations and reports endogenous violations and CAFA protein-centric $F_{\max}$ (Stage 0 / deletion / relaxation) over the Swiss-Prot panel of each species (1,200 proteins; 155 for *P. patens*).

Species (Clade)	Proteins (Inj.)	Stage 0 Viol.	Direct GO Taxon Viol.	Stage 1 Soft Viol.	Stage 2A Viol. (Del. / Rel.)	All-Protein $F_{\max}$: Stage 0 / Del. (Rec.) / Rel. (Rec.)	Injected $F_{\max}$: Stage 0 / Del. / Rel.	IC Retained	Dev. Rescue
S. cerevisiae (Fungi)	964 (300)	425	11 (97.4%)	30 (92.9%)	0/0 (100%)	0.6409 / 0.7101 (0.7510) / 0.6897 (0.7545)	0.6154/0.7165/0.6875	92.2%	100.0% (1/1)
C. elegans (Nematoda)	1,079 (300)	708	193 (72.7%)	98 (86.2%)	0/0 (100%)	0.4493 / 0.4847 (0.7382) / 0.4711 (0.7493)	0.4394/0.4887/0.4713	95.7%	0.0% (0/3)
D. melanogaster (Arthropoda)	1,081 (300)	941	279 (70.4%)	159 (83.1%)	0/0 (100%)	0.4726 / 0.4969 (0.6024) / 0.4924 (0.5555)	0.4692/0.4921/0.4883	94.2%	85.7% (12/14)
P. patens (Bryophyta)	147 (44)	55	10 (81.8%)	3 (94.5%)	0/0 (100%)	0.8099 / 0.8431 (0.9639) / 0.8291 (0.9669)	0.7752/0.8492/0.8189	99.8%	n/a (0/0)
A. thaliana (Magnoliopsida)	1,080 (300)	289	26 (91.0%)	15 (94.8%)	0/0 (100%)	0.6026 / 0.6273 (0.5791) / 0.6200 (0.5840)	0.5822/0.6429/0.6246	92.5%	100.0% (3/3)
M. musculus (Mammalia)	1,195 (300)	153	17 (88.9%)	12 (92.2%)	0/0 (100%)	0.8593 / 0.8632 (0.9399) / 0.8610 (0.9403)	0.8434/0.8601/0.8520	99.5%	92.0% (206/224)
Stress cohort total / mean	5,546 (1,544)	2,571	536 (79.2%)	317 (87.7%)	0/0 (100%)	0.6391 / 0.6709 (0.7624) / 0.6605 (0.7584)	0.6208/0.6749/0.6571	95.5%	90.6% (222/245)
Unperturbed cohort (mean of six species)	6,155 (0)	1,409	431 (69.4%)	236 (83.3%)	0/0 (100%)	0.7096/0.7134/0.7113	–	–	–

Prot proteins with the proteins' own species and genus excluded from the reference set, so that every annotation is a cross-species transfer. Stage 0 pooling of these claims produced 1,409 violations, concentrated in *C. elegans* and *D. melanogaster*, where conserved signaling and transcription-factor homologs import vertebrate organ-development terms onto invertebrate orthologs (Table 4). Direct GO taxon constraints removed 69% of them and left 431, because organ-level processes such as *heart development* (`GO:0007507`) carry their taxonomic restriction on the linked Uberon, PO, or CL class and not on the GO term, so detecting them requires traversing the logical definitions that connect a process to an anatomical structure. Stage 2A over the GO–Uberon/PO/CL bridge removed all 1,409, and six-species mean protein-centric $F_{\max}$ rose from 0.7096 to 0.7134 with deletion and 0.7113 with relaxation.

To compare the enforcement levels at a known error rate, we injected one cross-clade developmental claim into 300 randomly chosen proteins per species (44 in *P. patens*) on top of their real claims, and attenuated a fifth of the species-appropriate developmental claims scoring above 0.62 to test whether enforcement spares them (Figure 3c and Table 4). The injected cohort contained 2,571 violations after pooling; direct GO taxon rules left 536, the Stage 1 soft anatomical prior left 317, and Stage 2A enforcement left none. Attenuated species-appropriate claims exist in useful numbers only in the mouse, where enforcement with relaxation restored 206 of 224; the invertebrate and plant proteomes carry too few experimentally supported developmental annotations to test this. Enforcing anatomical validity in eukaryotes therefore requires the cross-ontology bridge, and the bridge retains valid developmental biology where it can be tested.

Once a term is inadmissible, Stage 2A can delete it or relax it to its nearest admissible ancestor. On real inputs, deletion gave the higher micro $F_{\max}$ in every species (six-species mean 0.671 against 0.661 over all cohort proteins and 0.675 against 0.657 on the injected proteins; Figure 3d), because a relaxed ancestor is often itself absent from the curated annotation and is then scored as a false positive; relaxation retained 95.5% of the information content of the pooled predictions and slightly more recall in four species. Deletion is therefore the better default for protein-centric accuracy, and relaxation is the option that keeps an annotation for a conserved homolog that cannot participate in *heart development* but may participate in *tube morphogenesis*, at a small cost in precision. Having examined constraints on complexes and admissible terms, we next asked how chromosomal context combines with metabolic network structure to locate missing enzymes.

2.5 Genomic context and protein language models locate missing pathway enzymes

Gap-filling identifies which reactions a network lacks and leaves open which gene catalyzes each one. Reaction-local genomic context (RLGC) scores candidate genes by their proximity and co-directionality relative to genes of reactions adjacent in the pathway, and we asked how much of a missing enzyme's identity this chromosomal signal recovers before any protein language model is involved. In 256 leave-reaction-out trials in *E. coli* K-12, where the held-out enzyme's own evidence was removed, ranking candidates without any target-specific evidence placed the true gene first in 2% of reactions, and METEOR's uniform reaction-level boost leaves such a ranking unchanged by construction. RLGC kernel density alone reached 23% unconditional Hit@1 (33% within the candidate pool), and adding the Stage 1 pathway coherence prior raised the within-pool value to 34% (Figure 4a and Table 5). Physical proximity to pathway neighbors on the chromosome therefore carries locus-specific information that reaction-level gap-filling discards.

Protein language models supply the complementary signal when reference enzymes for a reaction exist. Swiss-Prot ESM-2 reaction centroids, which in this track are not homology-masked, recovered 62% of the 256 enzymes on their own, and combining them with RLGC and coherence raised recovery to 64% (90% within the pool) by breaking ties among sequence-similar paralogs. The strictly masked configuration (Track A), in which both alignments and embedding neighbors above cosine 0.50 are removed, recovered 11%, which shows how much of the centroid signal is homology. Across the 414-reaction stratified benchmark, RLGC alone performed best on alternative catabolic pathways, whose enzymes and transporters sit in co-regulated operons, while full GSPA kept pool Hit@1 between 0.83 and 0.94 across core, catabolic, and peripheral strata. Under donor-homology masking (Protocol H), however, pool coverage fell to 27, 12, and 3 reactions at $\leq 70\%$, $\leq 50\%$, and $\leq 30\%$ identity and pool Hit@1 to 0.52, 0.58, and 0.33 (Figure 5c), so chromosomal context narrows the candidate set, and identifying a twilight-zone enzyme within it requires additional evidence. Transferred with unchanged parameters to 144 held-out reactions in *Bacillus subtilis* 168, full GSPA reached 43% unconditional and 82% pool Hit@1. To test the same mechanism on genomes without any curated annotation, we applied the pipeline to 100 high-quality metagenome-assembled genomes of uncultivated bacteria and archaea from the Empty Quarter desert, sampled across 17 phyla, with gaps defined as the reactions gapseq had to add to its draft model to reach growth (Figure 4d and Supplementary Table S11). Of 93 fillable gap reactions per genome, 12.6 received a gene at posterior ≥ 0.50 that Stage 0 had scored below 0.30, lifting MetaCyc pathway completeness of the annotation from 0.14 to 0.27 and essential-function completeness from 0.78 to 0.93. For 59% of these fills the chosen locus is the one gapseq's own sub-threshold search had hit without accepting, 39% name a locus gapseq had not seen, and reaction-local genomic context contributed only 18 fills, because with two evidence channels its scores rarely exceed the posterior of the alignment-based candidates. Gap-filling at this scale there-

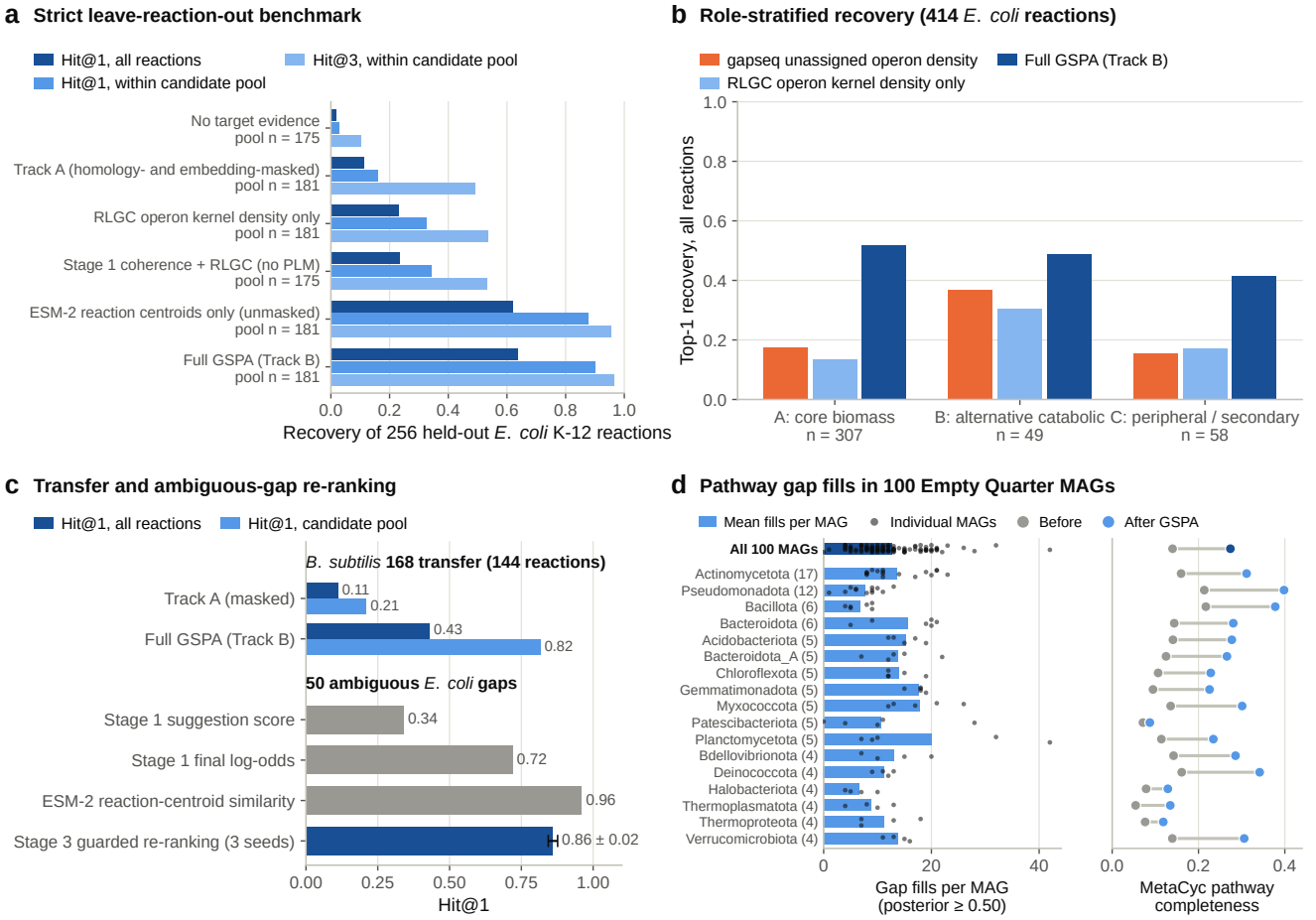


Figure 4. Leave-reaction-out metabolic enzyme attribution. (**a**) Recovery of 256 held-out reactions in *E. coli* K-12 (unconditional Hit@1, and Hit@1 and Hit@3 within the Stage 0 candidate pool, with pool size per method) for candidate ranking without target evidence, strict homology- and embedding-masked GSPA (Track A), RLGC kernel density alone, Stage 1 coherence with RLGC, ESM-2 reaction centroids alone, and full GSPA (Track B). (**b**) Unconditional Top-1 recovery across 414 *E. coli* reactions by role stratum (core biomass, alternative catabolic, peripheral or secondary) for gapseq operon density, RLGC alone, and full GSPA. (**c**) Out-of-sample transfer to *B. subtilis* 168 ($n = 144$) with parameters fixed on *E. coli*, and Top-1 recovery on the 50 ambiguous multi-candidate gaps for the single-feature rankings and the guarded Stage 3 agent. (**d**) High-confidence pathway gap fills per genome (posterior ≥ 0.50; bars are phylum means, points are genomes) and MetaCyc pathway completeness before and after refinement across 100 high-quality Empty Quarter metagenome-assembled genomes from 17 phyla.

fore mostly promotes alignment evidence that a metabolic model had already encountered, and the proposals remain ranked hypotheses for genomes that lack any experimental annotation.

Secondary metabolite clusters test the same chromosomal signal outside primary metabolism, where the target is a class label such as polyketide biosynthesis rather than a single reaction. Across the 1,292 antiSMASH regions in the 108 genomes of the invariant audit, the three Stage 0 channels attach a claim for the cluster's class term to 50% of core biosynthetic genes, but most of these claims are low-identity alignments calibrated below the acceptance threshold, so at posterior ≥ 0.50 class recall on core genes is 0.53 for siderophores, 0.16 for terpenes, 0.07 for non-ribosomal peptide synthetases, and near zero for polyketide synthases and for RiPPs, for which no channel emits the class term at all (Supplementary Table S10). The operon prior and the full Stage 1 and 2A configuration leave every value unchanged, because the genomic context prior propagates only functions that a KEGG pathway requires and the five class terms belong to no such pathway, while specificity on the 480,236 genes outside clusters stays at 0.9986. The chromosomal prior therefore acts on functions that a pathway table already names, and a cluster class that no pathway lists stays with the genes whose own evidence carries it. Whether a repaired reaction is stoichiometrically active is a separate question from which gene catalyzes it, and Stage 2B addresses it next.

2.6 Positive-cost flux repair attributes flux-carrying reactions to gene loci

The reaction-indicator MILP in METEOR assigns a negative cost to candidate reactions above an addition threshold, which rewards adding reactions that carry no flux. We quantified this effect on 108 bacterial genomes by auditing METEOR's archived repairs against a positive-cost threshold repair and against Stage 2B's two-step lexico-

Table 5. Leave-reaction-out (LRO) enzyme gene attribution in *E. coli* K-12, donor-homology masking (Protocol H), out-of-sample transfer to *B. subtilis* 168, and Stage 3 re-ranking of ambiguous gaps. *E. coli* K-12 served as the calibration genome. "All" scores every held-out reaction unconditionally, counting reactions whose true enzyme is absent from the Stage 0 candidate pool as misses; "Pool" scores only reactions whose true gene enters the pool (n_{cov}), so pool metrics are comparable only between rows with the same n_{cov}. In the first row the held-out enzyme carries no evidence of its own and candidates are ranked by the genome-scale suggestions alone; METEOR's uniform reaction-level boost leaves this ranking unchanged. Track B uses Swiss-Prot ESM-2 reaction centroids that are not homology-masked; Track A masks both alignments and embedding neighbors. For the 50 ambiguous gaps, the three single-feature rows rank the five candidates by one field of the evidence that the Stage 3 agent reads, and the Stage 3 row gives the mean ± population standard deviation over three seeds of the guarded agent (seed 42 at temperature 0.0, seeds 43 and 44 at 0.25) with de-identified prompts, in which every protein identifier is replaced by an opaque token (`results/pillar4_v2_multiseed_summary.tsv`).

Benchmark Cohort	Configuration	Reactions (n/n_{cov})	Hit@1 (All)	Hit@3 (All)	Hit@1 (Pool)	Hit@3 (Pool)	MRR (Pool)
E. coli K-12 strict LRO ($n = 256$)	No target evidence (candidate ranking from genome-scale suggestions)	256/175	1.95% (5)	7.03%	2.86%	10.29%	0.1354
	GSPA strict homology- and embedding-masked (Track A, cosine ≤ 0.50)	256/181	11.33% (29)	34.77%	16.02%	49.17%	0.3685
	RLGC operon kernel density only	256/181	23.05% (59)	37.89%	32.60%	53.59%	0.4745
	Stage 1 coherence + RLGC (no PLM)	256/175	23.44% (60)	36.33%	34.29%	53.14%	0.4864
	ESM-2 reaction centroids only (unmasked)	256/181	62.11% (159)	67.58%	87.85%	95.58%	0.9157
	Full GSPA (Track B: RLGC + coherence + PLM)	256/181	63.67% (163)	68.36%	90.06%	96.69%	0.9330
E. coli role-stratified & Protocol H ($n = 414$)	gapseq unassigned operon density (all strata)	414/414	19.57% (81)	34.78%	19.57%	34.78%	0.3054
	Full GSPA, Stratum A: core biomass	307/170	51.79% (159)	53.42%	93.53%	96.47%	0.9557
	Full GSPA, Stratum B: alternative catabolic	49/27	48.98% (24)	53.06%	88.89%	96.30%	0.9216
	Full GSPA, Stratum C: peripheral / secondary	58/29	41.38% (24)	48.28%	82.76%	96.55%	0.8793
	Full GSPA, all strata, Protocol H ≤ 70% masking	94/27	14.89% (14)	26.60%	51.85%	92.59%	0.6994
	Full GSPA, all strata, Protocol H ≤ 50% masking	57/12	12.28% (7)	17.54%	58.33%	83.33%	0.7125
	Full GSPA, all strata, Protocol H ≤ 30% masking	37/3	2.70% (1)	2.70%	33.33%	33.33%	0.4056
B. subtilis 168 out-of-sample LRO ($n = 144$)	GSPA strict homology-masked (Track A)	144/76	11.11% (16)	31.94%	21.05%	60.53%	0.4221
	Full GSPA (Track B, parameters fixed on *E. coli*)	144/76	43.06% (62)	50.00%	81.58%	94.74%	0.8933
Ambiguous multi-candidate LRO gaps ($n = 50$)	Stage 1 suggestion score alone	50/50	34.0% (17)	–	34.0%	–	0.5473
	Stage 1 final log-odds alone	50/50	72.0% (36)	–	72.0%	–	0.8450
	ESM-2 reaction-centroid similarity alone	50/50	96.0% (48)	–	96.0%	–	0.9767
	Stage 3 guarded re-ranking, de-identified prompts (3-seed mean ± s.d.)	50/50	86.00 ± 1.63%	–	86.00 ± 1.63%	–	0.9014

Table 6. Refinement of 148 RB-TnSeq fitness-validated bacterial enzymes (Price-149) across 22 host genomes starting from six enzyme predictors. Cells give correct Top-1 EC predictions (dataset-level macro $F_{\max}$ in parentheses) for each standalone predictor, after single-predictor METEOR v8 gap-filling (iteration 2, from archived METEOR repairs), after Stage 2B stoichiometric local-context probing (SLR + RLGC), after full single-predictor GSPA (iteration 3), after Stage 0 pooling, for the pool without the row's predictor, and after full multi-predictor GSPA (iteration 3). The pooled arm combines the row's predictor with a noisy-OR pool of CLEAN, DeepProZyme, MAPred, and EnzBERT; for these four predictors the baseline therefore enters twice (directly and inside the pool). The column "Pool without this predictor" removes the row's predictor and shows the pool of the other predictors only (for GraphEC and TopEC, which are not pooled, it is the four-predictor pool alone). The last column gives the Top-1 change from the standalone predictor to single-predictor and to multi-predictor GSPA, with the multi-predictor change in $F_{\max}$ in parentheses.

Predictor	Standalone	+ METEOR v8	+ Stage 2B SLR + RLGC	+ GSPA single-predictor	+ Stage 0 pooling	Pool without this predictor	+ GSPA multi-predictor	Single / multi gain
CLEAN (2023)	60 (0.4126)	61 (0.4048)	60 (0.4098)	63 (0.4216)	64 (0.4456)	54 (0.4596)	70 (0.4527)	+3 / +10 (+0.040)
MAPred (2025)	50 (0.3932)	52 (0.3985)	54 (0.3803)	56 (0.3704)	63 (0.4805)	69 (0.4731)	73 (0.5018)	+6 / +23 (+0.109)
DeepProZyme (2023)	49 (0.3869)	49 (0.3692)	51 (0.3853)	48 (0.4061)	59 (0.4669)	67 (0.4572)	60 (0.4587)	−1 / +11 (+0.072)
EnzBERT (2023)	53 (0.4488)	52 (0.4376)	53 (0.4468)	52 (0.4433)	65 (0.4730)	69 (0.4638)	62 (0.4841)	−1 / +9 (+0.035)
GraphEC (2024)	18 (0.2176)	18 (0.2216)	21 (0.2387)	20 (0.2164)	56 (0.4467)	68 (0.4738)	65 (0.4267)	+2 / +47 (+0.209)
TopEC (2025)	5 (0.0348)	5 (0.0353)	4 (0.0393)	5 (0.0390)	59 (0.4578)	68 (0.4738)	56 (0.4572)	+0 / +51 (+0.422)

graphic pFBA, all starting from the same DeepProZyme draft (Figure 5a and Supplementary Table S12). METEOR added a mean of 612 internal reactions per genome beyond the draft, of which 547 (89%) carried no flux under parsimonious FBA at the growth target; the positive-cost threshold repair added 53, of which fewer than one (1.6%) was flux-free; and the lexicographic pFBA reduced the active repair to 5.7 mass-balanced reactions per genome, every one of which carries flux. The repairs also differ in what they require of the medium: METEOR's and the threshold repair's networks reach the growth target only with uptake open on every exchange reaction, as in METEOR's own formulation, whereas Stage 2B solves on a defined minimal medium with a 36-compound supplement and reaches the target on it in all 108 genomes. A positive addition cost with lexicographic flux minimization therefore adds only reactions that participate in a flux-carrying, mass-balanced route on a stated medium, which is the property that competitive locus attribution then exploits. Because GSPA's draft core in this audit is seeded from the reactions METEOR had switched on, the comparison isolates the repair policy and not two independent reconstructions.

Whether attributing these reactions competitively to loci improves enzyme discovery was tested on 148 divergent bacterial enzymes from 22 genomes whose physiological roles were established by random-barcoded transposon mutagenesis (Price-149) [59], starting from six standalone enzyme predictors (Figure 5b and Table 6). Because 55 of the enzymes act in catabolic pathways that are inactive on minimal glucose, single-medium gap-filling lowered dataset-level $F_{\max}$ for four of the six predictors. Stage 2B's cognate-substrate probing, which rewards candidate reactions that share non-currency metabolites with chromosomal neighbors, recovered additional enzymes on a single predictor's output (MAPred 50 → 56 Top-1, CLEAN 60 → 63, GraphEC 18 → 21). The largest gains came from pooling CLEAN, DeepProZyme, MAPred, and EnzBERT in Stage 0 and then applying locus-level refinement: the pool of the three predictors other than MAPred identified 69 enzymes, and the pool with MAPred followed by refinement identified 73 (49% Top-1 accuracy, $p = 6 \times 10^{-6}$, exact McNemar test against MAPred alone), with dataset-level $F_{\max}$ rising from 0.393 to 0.502. The pool of the four predictors alone identified 68 enzymes, and adding GraphEC or TopEC to it lowered the result (65 and 56), so the apparent gains of those two predictors in the multi-predictor column come entirely from the pool; TopEC's

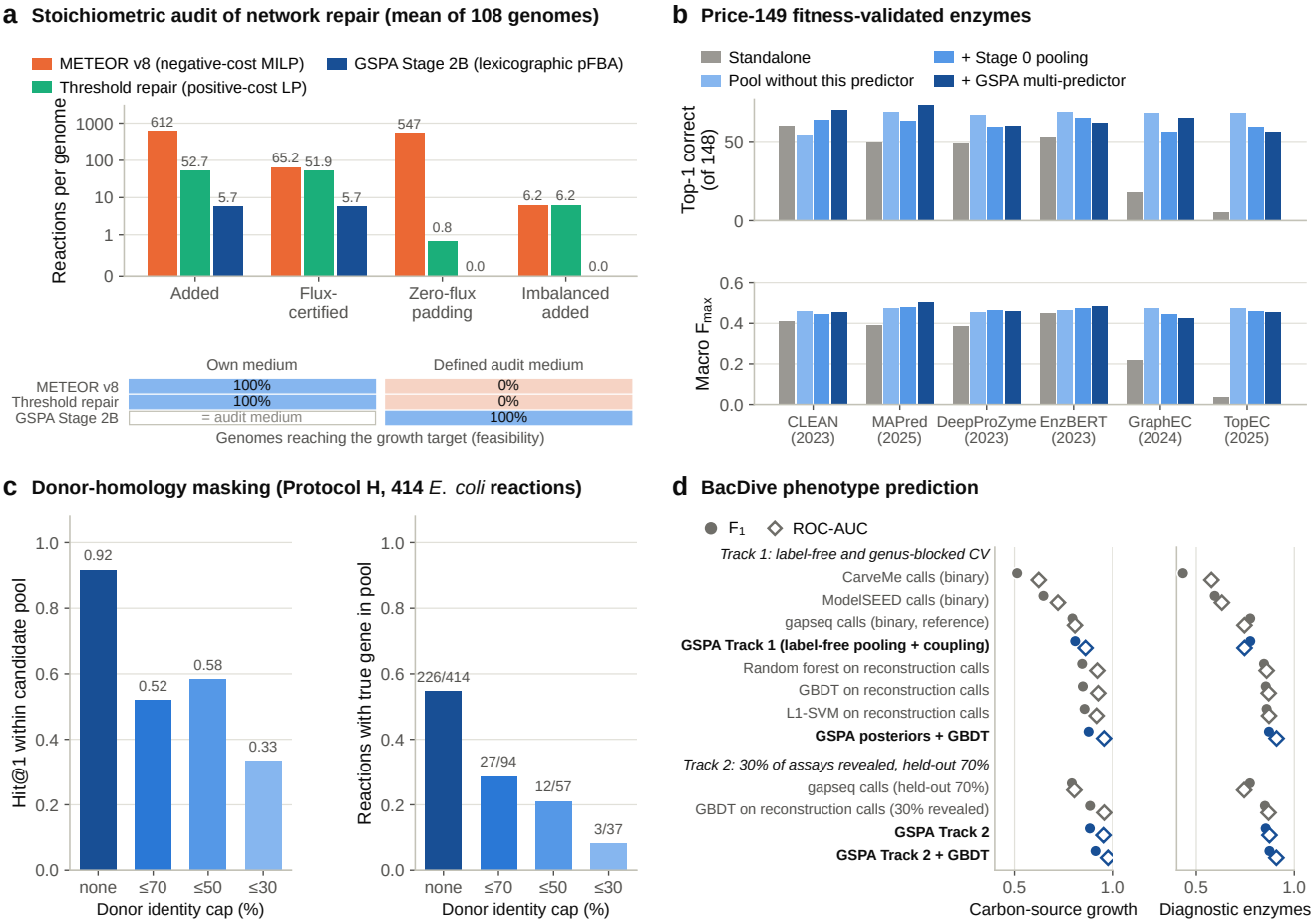


Figure 5. Stoichiometric audit, fitness-validated enzyme discovery, and experimental phenotype prediction. (**a**) Reactions added beyond the shared DeepProZyme draft, flux-certified additions ($|v_j| > 10^{-6}$ under pFBA), zero-flux additions, and elementally imbalanced additions per genome across 108 bacterial genomes for archived METEOR v8 repairs, positive-cost threshold repair, and Stage 2B lexicographic pFBA (logarithmic scale); the strip reports the fraction of genomes reaching the growth target on each method's own medium and on the defined audit medium. (**b**) Top-1 enzymes and dataset-level macro $F_{\max}$ on 148 RB-TnSeq fitness-validated enzymes (Price-149) for each standalone predictor, the pool of the other predictors, Stage 0 pooling, and full multi-predictor GSPA. (**c**) Pool Hit@1 and candidate-pool coverage for the 414 *E. coli* reactions under donor-homology masking at $\leq 70\%$, $\leq 50\%$, and $\leq 30\%$ identity. (**d**) F_1 and ROC-AUC on BacDive carbon-source growth and diagnostic enzyme assays for the reconstruction calls, GSPA Track 1 and Track 2, re-implemented genus-blocked classifiers, and the GSPA + GBDT hybrids.

released predictions also contain a single EC per protein, which limits its standalone score. Pooling and competitive stoichiometric attribution act additively on the strongest predictors, and the combination identifies fitness-validated enzymes that no predictor finds alone.

At the organism level, pathway posteriors should predict measurable physiology. We tested this on 2,332 carbon-source utilization assays from 791 strains and 17,496 diagnostic enzyme assays from 3,164 strains in BacDive [60, 61], using the binary pathway calls of gapseq, ModelSEED, and CarveMe as inputs (Figure 5d and Table 7). Track 1 pools the three reconstructions with a genus-peer consensus and an unsupervised cross-assay coupling term, all computed without phenotype labels, although its fixed weights were set by hand and no independent tuning set existed; it raised carbon-growth F_1 from 0.795 (gapseq) to 0.809 with better calibration (ROC-AUC 0.862 versus 0.808) and improved 43 assays while degrading 4 ($p = 3 \times 10^{-9}$), whereas on diagnostic enzymes it matched gapseq (0.776 versus 0.775). Under five-fold genus-blocked cross-validation, in which every test strain belongs to a genus absent from training, a gradient-boosted classifier on the GSPA posteriors reached $F_1 = 0.878$ on carbon growth and 0.871 on diagnostic enzymes, above re-implemented L1-SVM, random forest, and GBDT classifiers on the raw reconstruction calls (0.845–0.859). In Track 2, revealing 30% of a strain's assays and updating the remaining pathways through the coupling term lifted held-out carbon-growth F_1 from 0.792 to 0.884, and the hybrid reached 0.913. Coupled pathway posteriors transfer to unseen genera and propagate partial experimental observations to unassayed pathways. After these deterministic stages, a small set of loci remains ambiguous, which Stage 3 addresses.

Table 7. Experimental carbon-source growth and diagnostic enzyme phenotype prediction in BacDive. Inputs are the binary gapseq, ModelSEED, and CarveMe calls of the gapseq comparison tables [46]. Track 1 pools these calls with a genus-peer consensus and an unsupervised cross-assay coupling term without phenotype labels. Supervised rows are five-fold genus-blocked cross-validations (pooled out-of-fold metrics across unseen genera) of an L1-SVM, a random forest, and a gradient-boosted decision tree (GBDT) re-implemented on the reconstruction calls after the designs of Traitar, BacDive-AI, and DiASPora, and of a GBDT on the GSPA posteriors. Track 2 reveals 30% of each strainś assays as partial phenotypic evidence and evaluates the held-out 70%. The carbon-source table comprises 791 strains and 50 substrates and the enzyme table 3,164 strains after retaining assays with a definitive activity and calls from all three reconstructions. Middle rows report the 10-enzyme diagnostic panel (50,075 strain–EC pairs).

Assay Cohort & Protocol	Baseline or Method Pipeline	Strains / Assays	Precision	Recall	F_1 Score	Bal. Acc.	ROC-AUC	McNemar (+/−, p) or Runtime
Track 1: Carbon Growth (Blind & Genus-CV)	CarveMe calls (binary)	791/2,332	0.6935	0.4073	0.5132	0.6233	0.6233	+191/ − 597
	ModelSEED calls (binary)	791/2,332	0.8576	0.5200	0.6474	0.7214	0.7214	+186/ − 365
	gapseq calls (binary, reference)	791/2,332	0.8071	0.7836	0.7952	0.8082	0.8082	Reference
	GSPA Track 1 (label-free pooling + coupling)	791/2,332	0.8390	0.7818	0.8094	0.8239	0.8620	+43/ − 4 ($p = 2.8 \times 10^{-9}$)
	Random forest on reconstruction calls (5-fold genus-CV)	791/2,332	0.8433	0.8464	0.8448	0.8530	0.9222	+139/ − 37 ($p < 10^{-14}$)
	GBDT on reconstruction calls (5-fold genus-CV)	791/2,332	0.8475	0.8491	0.8483	0.8564	0.9268	+200/ − 90 ($p < 10^{-10}$)
	L1-SVM on reconstruction calls (5-fold genus-CV)	791/2,332	0.8472	0.8673	0.8571	0.8638	0.9185	+204/ − 78 ($p < 10^{-13}$)
	GSPA posteriors + GBDT (5-fold genus-CV)	791/2,332	0.8839	0.8718	0.8778	0.8848	0.9567	+291/ − 114 ($p < 10^{-18}$)
Track 1: Diagnostic Enzymes (Blind & Genus-CV)	CarveMe calls (binary)	3,164/17,496	0.7739	0.2996	0.4319	0.5774	0.5774	+939/ − 4,977
	ModelSEED calls (binary)	3,164/17,496	0.7854	0.4779	0.5943	0.6310	0.6310	+700/ − 3,263
	gapseq calls (binary, reference)	3,164/17,496	0.8393	0.7204	0.7753	0.7461	0.7461	Reference
	GSPA Track 1 (label-free pooling + coupling)	3,164/17,496	0.8425	0.7188	0.7757	0.7483	0.7465	+74/ − 52 ($p = 0.0609$)
	Random forest on reconstruction calls (5-fold genus-CV)	3,164/17,496	0.8391	0.8527	0.8458	0.7911	0.8593	+1,481/ − 317 ($p < 10^{-100}$)
	GBDT on reconstruction calls (5-fold genus-CV)	3,164/17,496	0.8602	0.8499	0.8550	0.8108	0.8699	+1,920/ − 509 ($p < 10^{-100}$)
	L1-SVM on reconstruction calls (5-fold genus-CV)	3,164/17,496	0.8402	0.8794	0.8593	0.8014	0.8715	+2,204/ − 790 ($p < 10^{-100}$)
	GSPA posteriors + GBDT (5-fold genus-CV)	3,164/17,496	0.8652	0.8774	0.8712	0.8256	0.9098	+2,568/ − 843 ($p < 10^{-100}$)
10-EC Diagnostic Panel (50,075 Assays)	DeepProZyme (2023) Standalone Baseline	10 ECs / 50,075 pairs	0.6442	0.8105	0.7178	0.6792	0.6792	Reference baseline
	+ METEOR a025 / v7 (Minimal-Glucose MILP)	10 ECs / 50,075 pairs	0.6285	0.8285	0.7148	0.6670	0.6670	Over-gap-filling drop
Track 2: Carbon Held-Out 70% (30% Revealed)	gapseq calls (binary)	791/1,641	0.8059	0.7789	0.7922	0.8049	0.8049	Reference
	GBDT on reconstruction calls (trained on 30% revealed)	791/1,641	0.8727	0.8985	0.8854	0.8901	0.9575	+170/ − 33 ($p < 10^{-22}$)
	GSPA Track 2 (Unsupervised Bayesian Anchor)	791/1,641	0.8290	0.9473	0.8842	0.8856	0.9536	+203/ − 78 ($p < 10^{-13}$)
	GSPA Track 2 + GBDT	791/1,641	0.9127	0.9139	0.9133	0.9175	0.9770	+241/ − 58 ($p < 10^{-26}$)
Track 2: Enzymes Held-Out 70% (30% Revealed)	gapseq calls (binary)	3,164/12,316	0.8376	0.7188	0.7737	0.7446	0.7446	Reference
	GSPA Track 2 (Unsupervised Bayesian Anchor)	3,164/12,316	0.8074	0.9078	0.8546	0.7755	0.8736	+1,707/ − 851 ($p < 10^{-64}$)
	GSPA Track 2 + GBDT	3,164/12,316	0.8622	0.8856	0.8737	0.8262	0.9088	+1,825/ − 564 ($p < 10^{-100}$)

Table 8. Stage 3 selective neuro-symbolic resolution on 80 de-identified anomalies and on 50 ambiguous leave-reaction-out gaps. The anomaly benchmark is built from real integrator output on seven genomes and contains 20 held-out pathway-gap true positives, 20 score-matched decoys, 20 injected cross-clade conflicts, and 20 valid borderline traps; protein and operon identifiers are replaced by opaque tokens. Precision, recall, F_1, and accuracy score the final action on each anomaly; promotions / relaxations count accepted promotions and ancestor relaxations; violations accepted counts promoted terms that are forbidden for the organism. The correct-gene column gives the percentage of the 20 true pathway gaps for which the correct gene was promoted. LRO-50 Hit@1 (MRR in parentheses) ranks the five candidate genes of each ambiguous leave-reaction-out gap; the three single-feature rows rank by one field of the evidence the agent reads and have no anomaly-benchmark entries (n/a). The language model is Qwen/Qwen3.8-27B-FP8 served by vLLM; the guarded multi-turn agent was run with seed 42 at temperature 0 and seeds 43 and 44 at temperature 0.25 (mean ± population s.d.; its LRO-50 entry is the three-seed mean), all other agent rows with seed 42 at temperature 0. Supervised classifiers are evaluated by five-fold stratified cross-validation; the best one per feature set (highest F_1, then accuracy) is shown. Bold marks the column maximum (minimum for violations accepted).

Configuration	Precision	Recall	F_1	Accuracy	Promotions / Relaxations	Violations Accepted	Correct Gene, 20 True Gaps	LRO-50 Hit@1 (MRR)
Deterministic Stages 1 + 2 only	0.0000	0.0000	0.0000	0.5000	0/0	0	0.0%	n/a
Direct zero-shot LLM (no tools, no guardrail)	0.0000	0.0000	0.0000	0.2750	37/0	1	0.0%	40.0% (0.6000)
Posterior threshold (≥ 0.35) + guardrail	0.0000	0.0000	0.0000	0.5000	40/20	0	0.0%	n/a
Best supervised, 3 pre-tool scalars + guardrail (GBDT)	0.1429	0.0500	0.0741	0.7250	7/0	0	5.0%	n/a
Best supervised, 8 post-tool features + guardrail (GBDT)	0.1667	0.0500	0.0769	0.7125	6/0	0	5.0%	n/a
Stage 3 fast 1-step	1.0000	0.1500	0.2609	0.6250	3/0	0	15.0%	36.0% (0.5800)
Stage 3 multi-turn without guardrail	0.5455	0.6000	0.5714	0.6000	22/4	1	60.0%	94.0% (0.9557)
Stage 3 multi-turn + guardrail (3-seed mean ± s.d.)	0.6177 ± 0.0866	0.5500 ± 0.0408	0.5811 ± 0.0614	0.7167 ± 0.0460	18.0/7.0	0	55.0 ± 4.1%	86.0 ± 1.6% (0.9014)
Single feature: suggestion score	n/a	n/a	n/a	n/a	n/a	n/a	n/a	34.0% (0.5473)
Single feature: Stage 1 final log-odds	n/a	n/a	n/a	n/a	n/a	n/a	n/a	72.0% (0.8450)
Single feature: ESM-2 reaction-centroid similarity	n/a	n/a	n/a	n/a	n/a	n/a	n/a	96.0% (0.9767)

2.7 Selective guarded language-model reasoning on ambiguous metabolic gaps

Stage 3 is triggered only by structured anomalies that survive Stages 1 and 2: unresolved pathway gaps, incomplete complexes, uncovered essential functions, and borderline ontology conflicts. On the 88 multi-kingdom proteomes these triggers selected 5.4% of proteins under the production cap of 150 anomalies per genome (14.7% uncapped; Supplementary Table S21), nearly all of them pathway gaps and borderline conflicts, which bounds the cost of invoking a language model. To evaluate the agent on anomalies the integrator itself produces, we held out one annotated enzyme at a time from pathways of seven genomes, kept the 50 cases in which the integrator then flagged the gap with the held-out protein among its candidates, and split them into 20 true positives and 20 decoys from which the held-out protein was removed entirely, matched on candidate count and maximum posterior; we added 20 injected cross-clade conflicts and 20 valid borderline terms as traps, all taken from the same extractor output with identical fields, protein identifiers replaced by opaque tokens, and candidate order shuffled (Figure 6 and Table 8). Incomplete complexes and uncovered essential functions could not be benchmarked this way because the extractor almost never places the held-out protein among the candidates (6 complex and 83 essential anomalies across all 88 proteomes). On these 80 items the guarded agent, equipped with five read-only tools for the operon neighborhood, the pathway gap, per-channel evidence, and ontology constraints, reached $F_1 = 0.58 \pm 0.06$ over three seeds (precision 0.62, recall 0.55) with no accepted constraint violation, chose the correct gene in 50–60% of the true gaps against 44% for an operon-context heuristic and 12.5% for chance, resolved all 20 injected conflicts, and kept 14 of the 20 traps. Removing the guardrail admitted a violation and lowered precision to 0.55; a single-turn mode promoted only 3 gaps; a zero-shot model without

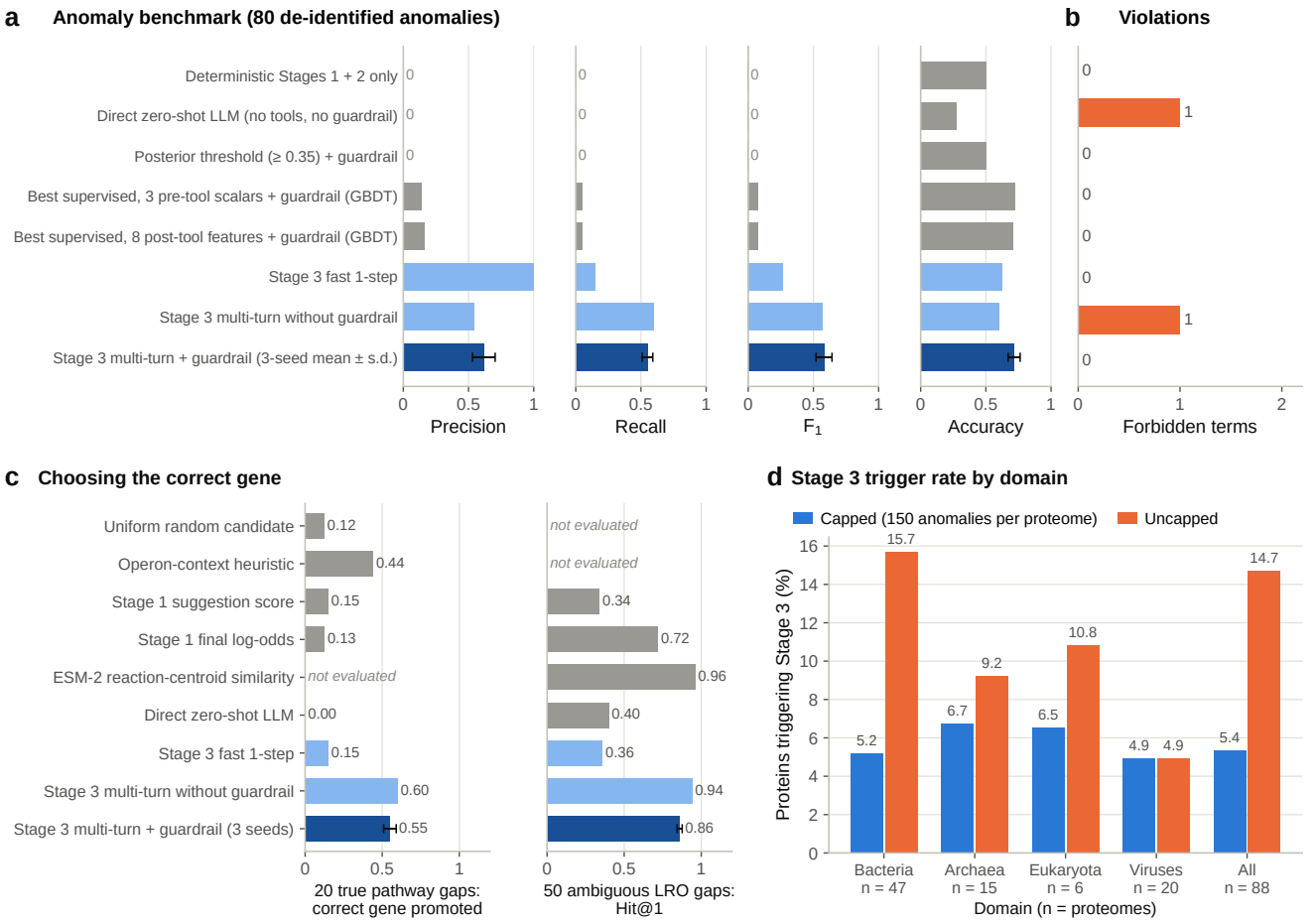


Figure 6. Evaluation of Stage 3 selective guarded reasoning. (**a**) Precision, recall, F_1, and accuracy on the 80 de-identified anomalies built from real integrator output (20 held-out pathway-gap true positives, 20 score-matched decoys, 20 injected cross-clade conflicts, and 20 valid borderline traps) for deterministic Stages 1 and 2, a zero-shot language model without tools, posterior thresholding with the guardrail, the best supervised classifiers on pre-tool and post-tool features with the guardrail, and the Stage 3 agent in single-step, unguarded, and guarded multi-turn modes (guarded: mean ± s.d. over three seeds). (**b**) Forbidden terms accepted by each configuration. (**c**) Fraction of the 20 true pathway gaps in which the correct gene was promoted, and Hit@1 on the 50 ambiguous leave-reaction-out gaps, including the single-feature rankings. (**d**) Fraction of proteins that trigger Stage 3 across the 88 multi-kingdom proteomes by domain, under the production cap of 150 anomalies per proteome and without it.

tools promoted nothing correctly; and supervised classifiers on the same tool features never selected a correct gap protein, so their accuracy of 0.75 reflects abstention and the guardrail rule on conflicts. The injected conflicts carry the integrator's own consistency penalty, which every method that reads prior contributions can exploit, so that category tests whether a method uses the pipeline's signal rather than whether it detects a conflict unaided. On the 50 ambiguous leave-reaction-out gaps in *E. coli* K-12 (Table 5), the agent placed the true enzyme first in $86 \pm 2\%$ of cases against 34% for the Stage 1 suggestion score, but the ESM-2 reaction-centroid similarity that it reads already reaches 96% on its own ($p = 0.07$ for the difference), so on that set the agent recovers most of one of its inputs and adds nothing beyond it. Guarded reasoning therefore resolves a useful fraction of real pathway gaps that deterministic inference leaves open and never violates the ontology, at a precision that still calls for human review.

2.8 Sequence dark matter across 72,533 bacterial genomes, and what whole-genome priors do at that scale

To see how much of the bacterial domain lies beyond reliable homology transfer, we aligned the coding sequences of 72,533 complete genomes from AllTheBacteria [62] (292 million coding sequences across 11,632 species and 37 phyla) against the experimentally annotated Swiss-Prot training set after deduplication and linear-time clustering (Supplementary Section S7 and Supplementary Table S16). Across all coding sequences, 41% had less than 30% identity to any training protein and a further 27% fell between 30% and 50%; 86% of the dark-matter fraction had no DIAMOND alignment at all at the sensitivity used. The dark fraction varied almost twofold across phyla, from 37% in Actinomycetota and 40% in Pseudomonadota to 61%–67% in environmental, predatory, and symbiotic phyla including Planctomycetota, Bacteroidota, Cyanobacteriota, and Spirochaetota, and it is higher still, about 50%,

in a set of 7,009 species-representative genomes from 29 phyla, because the collection as a whole is dominated by a few heavily sequenced pathogens. Two-thirds of bacterial proteins therefore lie in a regime where per-protein transfer is weak or impossible, which is the regime in which Table 1 shows the largest whole-genome gains.

We then ran the released integrator, with the same six priors and Stage 2A enforcement used throughout, over the 7,009 species-representative genomes and over 4,592 further genomes drawn from 230 batches, using the per-genome claims that the batch pipeline produces (Supplementary Section S7 and Supplementary Table S17). At this scale the priors changed the annotation status of 0.06% of proteins: 15,603 proteins gained a first annotation above 0.50, three-quarters of them through the essentiality prior and the rest through pathway and complex coherence and operon context, and 875,786 protein–term pairs were suppressed, 284,000 of them as taxon-forbidden terms and the remainder as genome-wide co-annotation conflicts; the gap-filling prior fired on no protein and no proteins without any alignment were annotated. Two properties of the batch pipeline explain this result, and both are limits of that deployment rather than of the priors measured in the earlier sections. The batch pipeline merges every Stage 0 channel into a single evidence source per claim, so the correlation-grouped pooling, the gap-filling prior, and the candidate pool that the coherence and operon priors act on receive one undifferentiated score per protein–term pair; and the bundled NCBI taxonomy covers the taxon of a third of these genomes, so taxonomic enforcement applies to the rest only through genome-wide co-annotation conflicts. Scaling whole-genome refinement to the bacterial domain therefore requires per-channel claims and full taxonomic coverage in the batch pipeline, and the alignment survey marks where those improvements would matter most.

3 Discussion

Protein function prediction has been formulated as a per-protein classification task in which each sequence or structure is scored independently [1, 21, 22, 25, 26, 28, 30]. GSPA reformulates proteome annotation as staged inference over the complete genome, and the evaluation separates three contributions. Calibrated pooling of homology, domain, convolutional, and language-model channels in Stage 0, and its stacked form GSPA-Seq, sets the per-protein baseline that every later stage builds on; whole-genome priors and cross-ontology enforcement in Stages 1 and 2A update that baseline with invariants over operons, essential functions, complexes, and admissible terms; and stoichiometric repair with competitive locus attribution in Stage 2B, together with selective guarded reasoning in Stage 3, connects network requirements to individual loci. On six withheld complete genomes the whole-genome stages improved weighted F-measures over GSPA-Seq on every time-based and homology split (Table 1), and the margin was largest on the homology splits, where alignment collapses and the retrained neural baselines lose most of their accuracy.

Stage 0 and GSPA-Seq follow the multi-source ensemble principle of GOLabeler [31] and NetGO [32, 33]: reliabilities are calibrated per channel, correlated channels are grouped so that shared homology or embedding evidence is counted once, and groups are pooled by Noisy-OR into an anchored log-odds prior. Because every Stage 1 update is added to that anchor, conserved proteins keep their high-confidence assignments and the recall of the Stage 0 candidate pool bounds what the priors can refine. The same dependence appears in leave-reaction-out attribution, where unconditional recovery is limited by whether the true gene enters the candidate pool at all, and in the homology splits, where the domain-signature channel stays almost flat because InterPro's models were built from the public proteomes that include our targets. Whole-genome inference adds organismal evidence on top of per-protein evidence.

What the whole-genome stages add is specific. Pre-t_0 annotations on neighboring genes of the same chromosome raised f_w on newly characterized genes by 0.056, and the ablations attribute the gain in Cellular Component mainly to enforcing taxonomic and anatomical admissibility before upward propagation, and the gain in Molecular Function mainly to pathway and complex coherence. Enforcement before propagation matters because a forbidden leaf term otherwise inflates shared intermediate ancestors that are scored as correct, and relaxation to the nearest admissible ancestor retains the conserved part of a cross-clade transfer, which is why it matched deletion on $F_{\max}$ while keeping more recall in tissue-forming eukaryotes (Table 4). Operon synteny raised top-ranked recovery of held-out complex subunits from 56% to 94% when the competing candidates lie outside the operon and operons are compact, far less in *Synechocystis*, whose complexes are dispersed, and not at all when the competitors are paralogs within the same operon, where only the Stage 3 agent separated them, and modestly (Table 3), so the prior's value is bounded by the genome's organization. Beyond homology, replacing METEOR's negative-cost reaction-indicator MILP [48] by positive-cost lexicographic pFBA [54] removed the zero-flux reactions that made up 89% of its additions, and competitive responsibility against an explicit unassigned-gene hypothesis, combined with predictor pooling, identified fitness-validated enzymes that no standalone predictor found (Table 6). At the organism level, coupled pathway posteriors predicted carbon-source growth and diagnostic enzyme activity in genera absent from training and propagated partial assay results to unassayed pathways (Table 7).

Several results delimit these claims. The whole-genome CAFA benchmark uses re-implemented external architectures trained on ESM-2 3B embeddings, so the external rows measure those architectures as we trained them, not the published models; the per-split recalibration of chan-

nel reliabilities on a validation set is part of why Stage 0 remains stable as homologs are removed, and the fairest comparison across splits is therefore the margin of full GSPA over GSPA-Seq, which is computed under identical calibration. The leave-subunit-out and perturbation tests use synthetic inputs by design, attenuated true subunits with synthetic decoys and injected cross-clade claims drawn from the constraint tables, so they establish that the priors act on the intended invariant at known error rates; synthetic decoys also carry no real claims for the complex role, which is why the real alignment channel alone ranks the true subunit first in most trials, and the unperturbed eukaryotic proteomes and the withheld-genome splits carry the evidence that such errors occur in practice. Leave-reaction-out recovery under donor-homology masking fell to between a third and a half of the candidate pool, which shows that chromosomal context narrows a candidate set but rarely identifies a twilight-zone enzyme by itself. For Stage 3, the benchmark built from the integrator's own anomalies covers only pathway gaps and ontology conflicts, because held-out complex subunits and essential functions rarely re-enter the candidate list, and on it the guarded agent is useful but imprecise ($F_1 = 0.58$); on the ambiguous leave-reaction-out gaps, the ESM-2 reaction-centroid similarity that the agent reads already resolves 96% of cases, so that set measures whether the agent recovers one of its inputs more than whether it reasons beyond them. Across the 72,533 AllTheBacteria genomes, the alignment survey places two-thirds of bacterial proteins in a regime of weak or absent homology, but the batch deployment of the integrator, which collapses the evidence channels into one source and covers less than half of the genomes' taxa, changed the annotation status of only 0.06% of proteins; the pan-phylum gains that the withheld-genome splits predict will only be realized once the batch pipeline carries per-channel claims and a complete taxonomy.

Three extensions follow directly from these limits. Stage 0 candidate recall can be raised by remote structural fold search and by protein language models trained under chronological splits, which would also allow a homology split that constrains every channel. Stage 2B models steady-state flux on specified media over reactions in the sanitized universal network, and Stage 2A enforces anatomy over the 2,715 GO terms that currently carry Uberon, PO, or CL logical definitions; condition-specific expression priors, transporter substrate prediction, and additional cross-ontology definitions would widen both. For Stage 3, a larger anomaly benchmark that includes complex and essential-function cases, evaluated leave-clade-out against supervised classifiers on the same tool features, would establish where guarded reasoning adds to deterministic inference, and heterologous expression of top-ranked candidates would turn the highest-confidence gap fills into tested hypotheses.

4 Methods

4.1 Problem formulation and four-stage architecture

Let $\mathcal{P} = \{p_1, \ldots, p_N\}$ denote the protein-coding genes of a target genome with NCBI Taxonomy identifier τ_{tax}, and let $\mathcal{K} = \{(c_i, s_i, e_i, \zeta_i)\}_{i=1}^{N}$ denote their genomic coordinates (contig c_i, start s_i, end e_i, and strand $\zeta_i \in \{+, -\}$). Let $\mathcal{T}$ denote the Gene Ontology (GO) directed acyclic graph under the true-path relation $t \sqsubseteq a$ (`is_a` and `part_of`), with ancestor closure $\text{Anc}(t)$, $\mathcal{M}$ the universal metabolic reaction network with stoichiometric matrix $\mathbf{S}$ and reaction–EC–GO mappings, and $\mathcal{O}$ the formal taxonomic and cross-ontology anatomical axiom set. GSPA estimates a proteome-wide posterior probability matrix $\mathbf{P} \in [0,1]^{N \times |\mathcal{T}|}$ (and corresponding EC probabilities in enzyme benchmarks) conditioned jointly on $\mathcal{K}$, τ_{tax}, $\mathcal{M}$, and $\mathcal{O}$ across four sequential stages (Figure 1): Stage 0 calibrates and pools per-protein evidence channels; Stage 1 performs iterative whole-proteome Bayesian log-odds refinement; Stage 2A applies deterministic symbolic constraint enforcement with semantic ancestor relaxation while Stage 2B performs positive-cost lexicographic parsimonious flux balance analysis (pFBA) and competitive gene-locus attribution; and Stage 3 applies selective tool-mediated neuro-symbolic reasoning verified by a symbolic ontology guardrail. Supplementary Table S1 lists every benchmark, its cohort, Stage 0 channels, active stages and command-line options, comparators, result files, and regeneration pipeline.

4.2 Stage 0: multi-channel evidence calibration and correlation-grouped Noisy-OR pooling

Stage 0 ingests functional claims from either a standalone predictor (such as CLEAN [27], MAPred [63], DeepProZyme [57], EnzBERT [56], GraphEC [64], or TopEC [65]) or up to seven complementary per-protein channels $e \in \mathcal{E}_{\text{ch}}$: (1) DIAMOND v2.1.8 sequence alignment against experimentally annotated Swiss-Prot [14]; (2) HMMER 3.4 profile HMM search against Pfam-A mapped via Pfam2GO [17]; (3) InterProScan v5.68-100.0 signature integration [18]; (4) ESM-2 (650M or 3B) embedding k-nearest-neighbor and multilayer perceptron transfer [23]; (5) ProtT5-XL-U50 embedding k-nearest-neighbor transfer [24]; (6) neural sequence classifiers (ProteInfer [66], 16-kernel DeepGOCNN and DeepGOPlus [20], DeepGOWeb [67], and DeepGO-SE [26]); and (7) structural similarity transfer via FoldSeek [29] and ProstT5 3Di structural tokens [68].

Each raw claim $c = (p, f, s(c), e(c))$ with raw score $s(c) \in [0,1]$ is mapped to a calibrated probability via Platt logistic scaling, $\tilde{q}(c) = \text{clip}\big(\sigma_{\log}(a_{e(c)}\, s(c) + b_{e(c)}),\, \varepsilon_{\text{cal}},\, 1 - \varepsilon_{\text{cal}}\big)$ with $\varepsilon_{\text{cal}} = 10^{-4}$, and weighted by channel reliability $r_{e(c)} \in (0,1]$ to give $q(c) = r_{e(c)}\, \tilde{q}(c)$ (channel pa-

rameters and reliabilities in Supplementary Section S1). To prevent double-counting correlated evidence sources, claims for each pair (p, f) are partitioned into correlation groups $\mathcal{G}_{\mathrm{corr}}(p, f)$ by evidence modality (homology, protein-sequence ML, structure, genomic context, genomic language models, inferred context, sequence motifs, localization, region features, and viral predictors, plus one group per domain-specific predictor; Supplementary Section S1). Stage 0 takes the maximum weighted probability within each group (with an optional second-opinion weight $\beta_{\mathrm{so}} = 0.0$ by default) and pools across independent groups via Noisy-OR:

$$q_g(p,f) = \mathrm{clip}\Big(\max_{c\in g}\big(r_{e(c)}\tilde{q}(c)\big) + \beta_{\mathrm{so}}\, q_{g,(2)}(p,f),\, 10^{-3},\, 0.999\Big),$$

$$P_0(p,f) = \mathrm{clip}\left(1 - \prod_{g\in\mathcal{G}_{\mathrm{corr}}(p,f)} \big(1 - q_g(p,f)\big),\, 10^{-6},\, 1-10^{-6}\right). \tag{1}$$

Stage 0 converts $P_0(p, f)$ into the anchored log-odds $L^{(0)}(p, f) = \mathrm{clip}\big(\log \frac{P_0(p,f)}{1-P_0(p,f)}, -12, 12\big)$ and propagates probabilities upward along GO `is_a` and `part_of` edges.

4.3 Stage 1: whole-proteome iterative Bayesian log-odds refinement

Stage 1 refines the proteome-wide probability matrix $\mathbf{P}^{(k)}$ over iterations $k = 1, \ldots, K_{\max}$ by adding six genome-scale prior log-odds shifts $B_m\big(p, f \mid \mathbf{P}^{(k-1)}, \mathcal{K}, \tau_{\mathrm{tax}}, \mathcal{M}, \mathcal{O}\big)$ to the fixed Stage 0 anchor $L^{(0)}(p, f)$ under Jacobi damping:

$$\tilde{L}^{(k)}(p,f) = \mathrm{clip}\left(L^{(0)}(p,f) + \sum_{m=1}^{6} w_m\, B_m\big(p, f \mid \mathbf{P}^{(k-1)}, \mathcal{K}, \tau_{\mathrm{tax}}, \mathcal{M}, \mathcal{O}\big),\, -12, 12\right),$$

$$L^{(k)}(p,f) = (1-\gamma_{\mathrm{iter}})L^{(k-1)}(p,f) + \gamma_{\mathrm{iter}}\tilde{L}^{(k)}(p,f), \qquad P^{(k)}(p,f) = \sigma_{\log}\big(L^{(k)}(p,f)\big), \tag{2}$$

with unit module weights $w_m = 1.0$, damping factor $\gamma_{\mathrm{iter}} = 0.50$, and maximum iterations $K_{\max} = 6$. Iteration terminates when the mean absolute probability change across updated entries falls below $\epsilon = 5 \times 10^{-3}$, and reverts to the last stable iterate if the mean change increases over two consecutive steps. The default values of all 18 configurable Stage 1 and Stage 2B hyperparameters (Supplementary Section S2 and Supplementary Table S3) were applied unchanged to every genome within each reported experiment. The six priors are defined as follows:

1. *Essentiality prior (B_1, $\alpha_{ess} = 1.50$):* Let $\mathcal{F}_{\mathrm{ess}}(\tau_{\mathrm{tax}})$ be the domain-specific core essential GO profile and $\mathcal{U}_{\mathrm{ess}}(\tau_{\mathrm{tax}})$ the essential categories with $\max_p P^{(0)}(p, e) \le 0.50$. Every claim on a term in the subclass closure of $\mathcal{F}_{\mathrm{ess}}$ receives a specificity-weighted boost of at most $0.30\,\alpha_{\mathrm{ess}}$, and for each uncovered category the single highest-ranked borderline candidate (likelihood in $[0.16, 0.50]$) receives a rescue floor $B_1(p, f) = \max\big(1.30\,\alpha_{\mathrm{ess}},\, 0.14 - L^{(0)}(p, f)\big)$ that lifts it to the decision threshold; no other claim is boosted.
2. *Pathway and heteromeric complex coherence prior (B_2, $\alpha_{coh} = 1.00$):* Missing steps of active GO `has_part` processes receive $w_{\mathrm{proc}}(f) = \max(0.25, s_{\mathrm{proc}}/n_{\mathrm{proc}})$, the fraction of satisfied steps, and missing steps of triggered reference pathways W with step coverage $c_W \ge 0.15$ receive $w_{\mathrm{path}}(f) = \max_{W\ni f}\big(0.35{+}0.65\,c_W\big)$. For an obligate heteromeric complex $f \in \mathcal{C}_{\mathrm{het}}$ with $N_{\mathrm{strong}}(f)$ proteins at $P^{(0)} > 0.50$ and $N_{\mathrm{cand}}(f)$ further candidates at $\max(P^{(0)}, P^{(k-1)}) \ge 0.08$, candidate partners receive $w_{\mathrm{cplx}}(p, f) = 0.85$ when $N_{\mathrm{strong}} = 1 \wedge N_{\mathrm{cand}} \ge 1$ (and 0.5525 when $N_{\mathrm{strong}} \ge 2$), whereas an isolated strong singleton ($N_{\mathrm{strong}} = 1 \wedge N_{\mathrm{cand}} = 0$) receives -0.75. The pathway and process terms act, like B_1, through a specificity-weighted boost and a rescue floor $\max\big(1.80\,\alpha_{\mathrm{coh}}\,w,\, 0.14 - L^{(0)}\big)$ on the highest-ranked borderline candidate (likelihood in $[0.18, 0.50]$) of each missing step, so that $B_2(p, f) = \alpha_{\mathrm{coh}}\big(\max(w_{\mathrm{proc}}(f), w_{\mathrm{path}}(f))\big)$ applies to that candidate and $\alpha_{\mathrm{coh}}\, w_{\mathrm{cplx}}(p, f)$ to every complex candidate.
3. *Taxonomic consistency prior (B_3, $\alpha_{cons} = 0.68$):* Active terms $\mathcal{A}^{(k-1)} = \{t \in \mathcal{T} : \max_p P^{(k-1)}(p, t) > 0.50\}$ are checked against direct GO taxon constraints and SAT4J unsatisfiable cores over multi-parent `is_a` and disjointness axioms. Any claim whose term or ancestor lies in the conflict set $\mathcal{V}_{\mathrm{conf}}^{(k-1)}$ receives $B_3(p, f) = -\alpha_{\mathrm{cons}}$.
4. *Anatomical prior (B_4, $\alpha_{anat} = 0.68$, $\alpha_{boost} = 0.65$):* Let $\mathcal{T}_{\mathrm{forb}}^{\mathrm{anat}}(\tau_{\mathrm{tax}})$ denote GO terms forbidden in τ_{tax} through the GO–Uberon/PO/CL bridge (Section 4.4) and $\mathcal{U}_{\mathrm{act}}(\tau_{\mathrm{tax}})$ valid non-root anatomical structures supported by at least two proteins at $P^{(0)} \ge 0.50$. Claims intersecting $\mathcal{T}_{\mathrm{forb}}^{\mathrm{anat}}(\tau_{\mathrm{tax}})$ receive $B_4(p, f) = -0.68$; valid claims linked to $\mathcal{U}_{\mathrm{act}}(\tau_{\mathrm{tax}})$ receive $+0.65$.
5. *Metabolic gapfilling and reaction-local genomic context prior (B_5, $\alpha_{gap} = 1.50$):* For each open metabolic network, the highest-ranked borderline candidate (likelihood in $[0.18, 0.50]$) for the gap's term receives $B_5(p, f) = \max\big(1.15\,\mu,\, 0.14 - L^{(0)}(p, f)\big)$ with $\mu = \alpha_{\mathrm{gap}}$ ($0.7\,\alpha_{\mathrm{gap}}$ for low-homology gapseq proposals), and all candidates for gap terms receive a specificity-weighted boost of at most $0.35\,\alpha_{\mathrm{gap}}$. For each gap reaction R, reaction-local genomic context (RLGC) identifies pathway-neighboring GO terms at 1–2 reaction hops ($\alpha_{\mathrm{rxn}} = 0.50$), weights chromosomal anchor genes p' with $P(p', g) \ge 0.30$ by $w_{\mathrm{anc}}(p', R) = \max_g \alpha_{\mathrm{rxn}}^{d(g)} P(p', g)$, and accumulates Gaussian kernel density $K_{\mathrm{bp}}(\Delta\mathrm{bp}) = \exp(-\Delta\mathrm{bp}^2/(2h_{\mathrm{bp}}^2))$ ($h_{\mathrm{bp}} =$ 5,000 bp):

$$D_{\mathrm{RLGC}}(p,R) = \sum_{p':|\Delta\mathrm{bp}(p,p')|\le 4h_{\mathrm{bp}}} w_{\mathrm{anc}}(p',R)\, K_{\mathrm{bp}}\big(\Delta\mathrm{bp}(p,p')\big),$$

$$S_{\mathrm{RLGC}}(p,R) = \log D_{\mathrm{RLGC}}(p,R) + \beta_{\mathrm{div}}\mathrm{Div}(p) - \gamma_{\mathrm{com}}\mathrm{Commit}(p) - \tag{3}$$

with neighbor-reaction diversity weight $\beta_{\mathrm{div}} = 1.0$, commitment penalty $\gamma_{\mathrm{com}} = 1.0$, self-annotation penalty $\delta_{\mathrm{self}} = 2.0$, same-strand bonus $\varepsilon_{\mathrm{str}} = 0.2$, and minimum-distance penalty $\zeta_{\mathrm{gap}} = 0.1$. In LRO enzyme attribution, S_{RLGC} is combined with orthogroup co-occurrence, Pfam domain compatibility, and (in Track

B) Swiss-Prot ESM-2 reaction-centroid cosine similarity.

6. *Co-directional operon prior (B_6, $\alpha_{ctx} = 1.00$, $\alpha_{gap_ctx} = 1.50$, $w_{op_cplx} = 0.95$):* Within a co-directional operon O ($|O| \geq 2$) having active pathway fraction $c_O(W) = k_O(W)/|O|$ for pathway W, a candidate (p, f) receives $B_6(p, f) = \max_{W \ni f : k_O(W) \geq 1} \left(\alpha_{\text{ctx}} c_O(W) \alpha_{\text{gap_ctx}}^{\mathbb{I}[f \in \mathcal{G}_{\text{gap}}]}\right) + w_{\text{op_cplx}} \mathbb{I}[f \in \mathcal{C}_{\text{het}} \wedge \exists p' \in O \setminus \{p\} : P^{(k-1)}(p', f) > 0.50]$.

The pathway-consensus part of B_6 acts in the same targeted form (rescue floor $\max(1.65\,\alpha_{\text{ctx}}, 0.14 - L^{(0)})$ on the highest-ranked borderline candidate of each operon-supported pathway, specificity boost of at most $0.38\,\alpha_{\text{ctx}}$), while its complex-partner term applies to every co-operonic candidate. The rescue floors, their candidate windows, and the ranking rule are listed with their source modules in Supplementary Section S2.

4.4 Cross-ontology taxonomic and anatomical constraint bridge

To enforce anatomical and taxonomic consistency across domains, we extracted five programmatic constraint tables from the official OBO Foundry releases of GO, Uberon [10], Plant Ontology (PO) [50], Cell Ontology (CL) [51], and NCBI Taxonomy (Supplementary Section S3 and Supplementary Table S7): (1) 26,911 direct GO taxon constraints (13,819 `only_in_taxon` and 13,092 `never_in_taxon`) across 18,159 GO terms [9]; (2) 2,715 cross-ontology equivalence and subclass axioms linking 2,674 GO terms to 1,294 Uberon, PO, and CL classes across 30 relations (including `results_in_development_of`, `results_in_morphogenesis_of`, `results_in_formation_of`, `results_in_maturation_of`, and `occurs_in`); (3) 8,331 taxon constraints (4,767 `only_in_taxon` and 3,564 `never_in_taxon`) over 1,247 Uberon, PO, and CL anatomical classes; (4) 7,722 transitive `is_a` and `part_of` edges over 2,855 anatomical classes; and (5) 1,818 taxonomic lineage edges over 1,305 NCBI Taxonomy clades. A GO term t is entailed as anatomically forbidden in taxon τ_{tax} whenever t (or a GO ancestor of t) links to an anatomical class u whose anatomical ancestor closure carries an `only_in_taxon` or `never_in_taxon` rule incompatible with the taxonomic lineage of τ_{tax}.

4.5 Stage 2A: deterministic symbolic enforcement and semantic ancestor relaxation

After Stage 1 convergence, Stage 2A deterministically enforces all direct GO taxon constraints, cross-ontology anatomical constraints, and SAT4J logical consistency axioms prior to final upward GO ancestor closure, ensuring that clade-incompatible leaf claims are resolved before parent terms are populated. For each protein p and forbidden term t, Stage 2A traverses $\text{Anc}(t)$ (excluding the three GO aspect roots) to identify the minimal valid ancestor set $\mathcal{A}_{\text{valid}}(t, \tau_{\text{tax}})$ comprising the most specific ancestors of t that are both taxonomically and anatomically admissible in τ_{tax}. Reconstructing the unpenalized pre-conflict probability $P_{\text{unpen}}(p, t)$ from $L^{(0)}(p, t)$ and positive Stage 1 priors, semantic ancestor relaxation transfers this support to each minimal valid ancestor and zeroes the forbidden term:

$$P_{\text{Stage2A}}(p, a) = \max\Big(P^{(K)}(p, a),\, \gamma_{\text{relax}}\, P_{\text{unpen}}(p, t)\Big), \qquad \forall a \in \mathcal{A}_{\text{valid}}$$
$$P_{\text{Stage2A}}(p, t) = 0, \tag{4}$$

with $\gamma_{\text{relax}} = 1.0$, followed by upward true-path ancestor closure over admissible terms. In addition, for any obligate heteromeric complex $f \in \mathcal{C}_{\text{het}}$ with a single protein at $P \geq 0.50$ and zero partner candidates at $P \geq 0.08$ across the genome, Stage 2A attenuates the isolated singleton to 0.45.

4.6 Stage 2B: role-aware reaction support and positive-cost lexicographic pFBA

Stage 2B is implemented as a Python module (`stoichiometric_locus_refiner.py`, SciPy with the HiGHS solver) that the JVM pipeline invokes, with its ModelSEED-derived matrices distributed as a separate data bundle. It operates over a sanitized universal metabolic network $\mathcal{U}_{\text{clean}}$ derived from ModelSEED [44] (47,880 reactions and 33,978 metabolites, of which 14,243 carry EC mappings and 11,826 are mass- and charge-balanced enzymatic reactions; 2,417 pool or elementally imbalanced reactions receive penalty $w_{\text{imb}}(j) = 100.0$). Each reaction j is associated with a disjunctive normal form (DNF) catalytic rule $\mathcal{M}_{\text{dnf}}(j) = \{m_1, \ldots, m_{K_j}\}$ over alternative isozyme mechanisms m, where each mechanism requires a conjunction of subunit roles $\mathcal{R}_{\text{sub}}(m) = \{u_1, \ldots, u_{D_m}\}$. Given candidate locus probabilities $s_{p,u} \in (0, 1)$, Stage 2B aggregates the top $M_{\text{iso}} = 3$ candidates per role via Noisy-OR, combines required subunits via a bottleneck-gated geometric mean at $\theta_{\text{sub}} = 0.25$, and takes the maximum across isozyme mechanisms:

$$Q(u) = 1 - \prod_{p \in \text{Top-}M_{\text{iso}}(\mathcal{C}_{\text{cand}}(u))} \big(1 - s_{p,u}\big),$$
$$S_{\text{mech}}(m) = \begin{cases} \Big(\prod_{u \in \mathcal{R}_{\text{sub}}(m)} Q(u)\Big)^{1/D_m}, & \min_u Q(u) \geq \theta_{\text{sub}}, \\ \min_{u \in \mathcal{R}_{\text{sub}}(m)} Q(u), & \text{otherwise}, \end{cases} \qquad S_{\text{rxn}}(j) \tag{5}$$

Reactions with $S_{\text{rxn}}(j) \geq \tau_{\text{core}} = 0.50$, together with open medium exchanges and spontaneous reactions, form the draft core $\mathcal{R}_{\text{core}}$. Candidate repair reactions $j \in \mathcal{D}_{\text{dom}} \setminus \mathcal{R}_{\text{core}}$ receive a strictly positive addition cost $c_j^{\text{add}} = \lambda_{\text{add}}(1 - S_{\text{rxn}}(j)) + w_{\text{imb}}(j) + \epsilon_{\text{par}}$ ($\lambda_{\text{add}} = 1.0$, $\epsilon_{\text{par}} = 10^{-3}$). Splitting reversible fluxes into non-negative

components $v_j = v_j^+ - v_j^-$ and setting the biomass target on medium $\mathcal{E}_{\text{med}}$ to $v_{\text{bio}}^{\text{target}}(\mathcal{E}_{\text{med}}) = \max(\alpha_{\text{bio}} v_{\text{bio}}^{\max}(\mathcal{E}_{\text{med}}), 0.1)$ ($\alpha_{\text{bio}} = 0.10$), Stage 2B solves a two-step lexicographic pFBA [54] using the HiGHS linear programming solver (feasibility tolerance 10^{-9}). Step 1 minimizes the repair cost $\sum_{j \notin \mathcal{R}_{\text{core}}} c_j^{\text{add}}(v_j^+ + v_j^-)$ subject to $\mathbf{Sv} = \mathbf{0}$ and $v_{\text{bio}} \geq v_{\text{bio}}^{\text{target}}(\mathcal{E}_{\text{med}})$ (solving first over balanced reactions and admitting penalized imbalanced reactions only if infeasible) to obtain optimal repair cost $C^*(\mathcal{E}_{\text{med}})$. Step 2 minimizes total non-biomass flux $\sum_{j \neq \text{bio}}(v_j^+ + v_j^-)$ subject to $\sum_{j \notin \mathcal{R}_{\text{core}}} c_j^{\text{add}}(v_j^+ + v_j^-) \leq (1 + 10^{-6}) C^*(\mathcal{E}_{\text{med}}) + 10^{-9}$. The active repair set $\mathcal{J}_{\text{rep}}(\mathcal{E}_{\text{med}}) = \{j \notin \mathcal{R}_{\text{core}} : |v_j^{\text{pFBA}}(\mathcal{E}_{\text{med}})| > 10^{-6}\}$ is evaluated by flux-variability ($1.01C^*$), reaction-knockout ($1.10C^*$), and five-replicate cost-perturbation ($\pm 5\%$) certificates to yield the flux confidence $\pi_j^{\text{FBA}} = 0.5\mathbb{I}[j \in \mathcal{R}_{\text{FVA}}] + 0.3\mathbb{I}[j \in \mathcal{R}_{\text{KO}}] + 0.2\omega_{\text{ens}}(j) \in [0, 1]$ (Supplementary Section S4 and Supplementary Table S12).

4.7 Stage 2B: competitive gene-locus responsibility and substrate probing

For each repaired reaction $j \in \mathcal{J}_{\text{rep}}$ and required role $u \in \mathcal{R}_{\text{sub}}(m_j^*)$, Stage 2B distributes locus responsibility competitively across candidate genes $p \in \mathcal{C}_{\text{cand}}(u)$ and an explicit unassigned-gene hypothesis $h_\varnothing$ with prior mass $\eta_{\text{null}} = 0.05$:

$$w_{j,u}(p) = \big(o_{p,u}\big)^{\beta_1} \, \text{BF}_{\text{operon}}(p,u)^{\beta_2} \, \exp\big(\beta_3 C_{\text{syn}}(p,u) + \beta_4 S_{\text{ESM2}}(p,u)\big),$$
$$r_{j,u}(p) = \frac{w_{j,u}(p)}{\eta_{\text{null}} + \sum_{p' \in \mathcal{C}_{\text{cand}}(u)} w_{j,u}(p')}, \tag{6}$$

where $o_{p,u} = s_{p,u}/(1 - s_{p,u})$ is the sequence odds, $\text{BF}_{\text{operon}}(p,u)$ is the RLGC operon Bayes factor, $C_{\text{syn}}(p,u)$ is the operon and pathway synteny score, $S_{\text{ESM2}}(p,u)$ is the language-model similarity score, and $\beta_1 = \beta_2 = 1.0, \beta_3 = \beta_4 = 1.5$. The backward log-odds update to locus p is $\Delta L_{\text{FBA}}(p,u) = \gamma_{\text{FBA}} \sum_{j \in \mathcal{J}_{\text{rep}}(u)} \pi_j^{\text{FBA}} r_{j,u}(p)(1 - S_{\text{rxn}}(j))$ with $\gamma_{\text{FBA}} = 2.20$, applied across damped outer iterations $\lambda_t = \lambda_0 \gamma_{\text{damp}}^{t-1}$ ($\lambda_0 = 0.70, \gamma_{\text{damp}} = 0.60$). For catabolic enzymes in Price-149, Stage 2B supplements minimal-glucose pFBA with cognate-substrate stoichiometric neighborhood probing (rewarding candidate reactions that share non-currency metabolites at 1–2 reaction hops with chromosomal neighbors within ± 10 genes) and blends single-predictor and five-predictor ensemble posteriors (complete equations in Supplementary Section S4). Held-out metabolic reactions are classified by five binary recoverability indicators: structural presence and mass balance in $\mathcal{U}_{\text{clean}}$ ($\mathcal{C}_{\text{struct}}$), medium-specific flux demand ($\mathcal{C}_{\text{task}}$, partitioning core biomass Stratum A, cognate-substrate catabolic Stratum B, and secondary Stratum C), isozyme uniqueness ($\mathcal{C}_{\text{unique}}$), operon support ($\mathcal{C}_{\text{operon}}$), and multi-subunit partner presence ($\mathcal{C}_{\text{complex}}$).

4.8 Stage 3: selective neuro-symbolic annotation completion

Stage 3 is triggered selectively on four post-Stage-2 anomaly categories: (1) pathway gaps (pathway completeness ≥ 0.50 with candidate posteriors in $[0.05, 0.50)$); (2) incomplete obligate heteromeric complexes (one active subunit at $P \geq 0.50$ and sub-threshold partner candidates); (3) borderline uncovered essential categories; and (4) borderline taxonomic or anatomical conflicts ($P \in [0.35, 0.62]$) amenable to ancestral GO relaxation. Stage 3 evaluates at most 8 candidates per anomaly and 150 anomalies per genome. The Stage 3 benchmark was built from the extractor's own records: for each of seven genomes (*E. coli* K-12, *B. subtilis*, *P. aeruginosa*, *M. tuberculosis*, *S. cerevisiae*, *C. elegans*, and *A. thaliana*) every annotated pathway step was held out in turn by deleting the protein's claims for the step's term and its descendants, the integrator was re-run with the anomaly cap lifted, and the cases in which the held-out protein re-entered the candidate list were split by hash into true positives and decoys (held-out protein removed from the genome and the operon table), matched on candidate count and maximum posterior within ± 0.05; conflicts were injected as cross-clade terms at raw scores on a grid from 0.40 to 0.85, chosen so that their Stage 1 posteriors fell in the borderline band $[0.35, 0.50)$, into Stage 1 runs without Stage 2A, and traps are valid borderline terms from the same runs. All GO labels were taken from the GO release of 2026-03-25, and 8 records whose target term is absent from the guardrail's GO release were excluded. Supervised baselines (logistic regression, random forest, GBDT, and MLP) were evaluated by five-fold stratified cross-validation on three pre-tool scalars or eight post-tool features, with and without the guardrail.

Each anomaly is stripped of evaluation labels and gene symbols and processed in a bounded multi-turn tool-calling loop (maximum 6 turns; 2 turns in the fast mode) by a locally hosted FP8-quantized open-weight instruction model (`Qwen/Qwen3.8-27B-FP8`) served through vLLM on two NVIDIA RTX 4090 GPUs (131,072-token context, maximum 450 output tokens per turn; seed 42 at temperature 0.0 and seeds 43 and 44 at temperature 0.25). The model invokes five deterministic read-only tools to inspect the operon neighborhood, inspect the pathway or complex gap, query protein evidence and domain signatures, check symbolic ontology constraints, and propose a structured resolution (PROMOTE, RELAX, DEMOTE, or ABSTAIN). Every proposal is verified by the deterministic symbolic guardrail against the 48,101 GO term stanzas, 1,305 NCBI Taxonomy clades, 26,911 direct GO taxon rules, and 2,715 GO–Uberon/PO/CL axioms before acceptance (tool schemas, de-identification protocol, per-seed results, and the trigger-rate computation in Supplementary Section S6 and Supplementary Tables S14, S20, and S21).

4.9 Experimental phenotype prediction in BacDive

We evaluated organism-level phenotype prediction on the two comparison tables published with gapseq [46]: 2,332 ProTraits carbon-source utilization assays (791 strains, 50 substrates; all rows retained) and 17,894 BacDive enzyme assays [60, 61], of which we retained the 17,496 assays (3,164 strains) with a definitive activity (+ or −) and calls from all three reconstructions, and on 50,075 strain–EC pairs for 10 diagnostic enzymes from per-EC confusion matrices. Binary pathway or reaction calls $g(s,y), m(s,y), c(s,y) \in \{0,1\}$ of gapseq [46], ModelSEED [44], and CarveMe [45] for strain s and assay y were taken from the comparison tables published with gapseq. We form the blended matrix $B(s,y) = 0.50g + 0.32m + 0.18c$ (carbon; $0.52g+0.30m+0.18c$ for enzymes), column-center each assay $\tilde{B}_{s,y} = B(s,y) - \bar{B}_y$, and compute the row-normalized non-negative cosine coupling matrix $\Psi(y,y')$ with zero diagonal to obtain the unsupervised cross-assay coupling score $\psi(s,y) = \sum_{y'} \Psi(y,y')(B(s,y') - 0.5)$. In Track 1, which uses no phenotype labels, binary calls are mapped to fixed probabilities ($p_{\text{gap}} = 0.80/0.22$, $p_{\text{ms}} = 0.78/0.24$, $p_{\text{cm}} = 0.68/0.34$ for positive/negative calls), combined with the Laplace-smoothed mean blended call p_{peer} of the other strains of the same genus, and pooled in log-odds space:

$$P_{\text{Track1}}(s,y) = \sigma_{\log}\Big(w_{\text{gap}}\,\text{logit}\big(p_{\text{gap}}(s,y)\big) + w_{\text{ms}}\,\text{logit}\big(p_{\text{ms}}(s,y)\big) + w_{16}\,\text{logit}\big(p_{\text{cm}}(s,y)\big) + w_{\text{peer}}\,\text{logit}\big(p_{\text{peer}}(s,y)\big) + w_{\psi}\,\psi(s,y)\Big), \tag{7}$$

with weights $(w_{\text{gap}}, w_{\text{ms}}, w_{\text{cm}}, w_{\text{peer}}, w_{\psi}) = (0.42, 0.26, 0.12, 0.24, 0.85)$ for carbon and $(0.42, 0.26, 0.12, 0.22, 0.85)$ for enzymes at threshold 0.50; these weights were fixed by hand, without a tuning set separate from the evaluated assays. Supervised baselines were re-implemented as an L_1-regularized linear support vector machine, a random forest, and a gradient-boosted decision tree over the three binary calls and a one-hot assay indicator, after the designs of Traitar [69], BacDive-AI [61], and DiASPora [70]; they and the GSPA + GBDT hybrid were evaluated under five-fold genus-blocked cross-validation with all calibration, prevalence, and coupling parameters estimated strictly within training folds. In Track 2, 30% of each strain's assays are revealed to condition the Bayesian anchor $P_{\text{Track2}}(s,y)$ on revealed substrate co-occurrence or genus-level EC posteriors, and predictions are evaluated on the held-out 70% (complete parameterization in Supplementary Section S5 and Supplementary Table S13).

4.10 Benchmarks, time-based and homology-based split retraining, metrics, and structural audit score

To evaluate prospective generalization across time and across sequence-similarity regimes on complete genome inputs, we constructed a six-split whole-genome withheld CAFA benchmark across six complete prokaryotic reference proteomes (*B. subtilis* 168, *P. aeruginosa* PAO1, *M. tuberculosis* H37Rv, *H. pylori* 26695, *Synechocystis* sp. PCC 6803, and *M. jannaschii* DSM 2661; 20,697 coding sequences, 3,315 experimentally annotated target proteins, 1,060 strictly post-t_0 targets, 3,809 operons, and 1,467 metabolic gaps) alongside a multi-clade isolated-protein CAFA benchmark (Table 1 and Supplementary Tables S4 and S15). In the two whole-genome time-based splits, post-t_0 targets (1,060 proteins) are evaluated either with pre-t_0 experimental annotations on the six chromosomes retained in training to test chromosomal rollover context (`WG_Time_Rollover`, 82,404 training proteins) or with all 2,524 experimentally annotated proteins from the six species purged from training (`WG_Time_Withheld`, 79,880 training proteins). In the four whole-genome homology splits, all six species are purged from training and evaluated across all 3,315 experimental target proteins after additionally purging all cross-species training homologs sharing pairwise sequence identity $> \eta_{\text{id}}$ with any protein in the six withheld genomes (`WG_Homology_le100`: 79,880 train; `WG_Homology_le70`: 79,457 train; `WG_Homology_le50`: 76,405 train; `WG_Homology_le30`: 61,323 train). On the two time-based splits, we additionally evaluate the official pre-trained weights of DeepFRI [28] and DeepGO-SE [26]. On every split, all 14 trainable per-protein baselines (term-frequency prior, filtered DIAMOND database, L_2-regularized InterPro domain logistic regression, canonical and reranking convolutional DeepGOCNN with $\{8, 16, \ldots, 128\} \times 512$ filters = 8,192 features, DeepGOPlus [20], 2,560-dimensional ESM-2 3B `esm2_t36_3B_UR50D` k-NN index [71], ESM-2 3B MLP, hierarchical ESM-2 3B C-HMCNN with max-constraint loss [72], DeepFRI graph convolutional and language-model architecture [28], DeepGraphGO two-layer graph convolutional network over the DIAMOND sequence-similarity graph with InterPro node features [21], TALE+ transformer encoder with hierarchical GO label-embedding joint attention and DIAMOND fusion [22], SPROF-GO language-model self-attention pooling with hierarchical label diffusion and homology transfer [25], DeepGO-SE five-model approximate semantic entailment ensemble over $\mathcal{EL}^{++}$ description-logic embeddings [26], and DPFunc InterPro domain cross-attention with graph convolutional layers [30]) together with our out-of-fold GSPA-Seq meta-stacker are retrained on that split's filtered training set before running whole-genome inference across all 20,697 coding sequences. In this benchmark, Stage 0 pooling and the Stage 1, 2A, and 2B updates are implemented in Python inside the benchmark script (`scripts/run_cafa_whole_genome_splits.py`), following the rules above on the same operon, gap, pathway, complex, and constraint tables, so that each split can be evaluated with its own retrained channels; all other benchmarks run the released JVM integrator. The external architectures are our re-implementations: all take ESM-2 3B embeddings as sequence input, DeepFRI uses sequence-band graph smoothing in place of contact maps,

DeepGraphGO uses the DIAMOND similarity graph in place of a protein interaction network, and sequences are truncated to 256 residues for the DeepFRI and TALE+ encoders. Stage 0 channel reliabilities are recalibrated on a 600-protein validation set drawn from each split's training set (aligned with DIAMOND against the same training database), and a common down-weighting factor for the convolutional and ESM-2 channels is selected from $\{1.0, 0.75, 0.55, 0.38\}$ by validation f_w on that split (0.75 on the time-based and $\leq 100\%$ splits, 1.0 otherwise). For the pan-phylum AllTheBacteria and reference eukaryote analysis (72,533 complete genomes across 11,632 species and 37 phyla [62]; Supplementary Section S7 and Supplementary Table S16), coding sequences across 73 assembly batches (315,671,487 proteins) were deduplicated by SHA-256 sequence hashing (73,866,871 unique sequences) and clustered via `MMseqs2 linclust` [73] at 90% sequence identity and 80% coverage (36,804,001 non-redundant cluster representatives), and aligned with DIAMOND (`-sensitive`, $k = 25$, $E \leq 10^{-3}$) against the Swiss-Prot training set to obtain the per-protein maximum identity reported in Section 2.8. The batch pipeline then scores each representative with DIAMOND transfer, Swiss-Prot-derived domain annotations, and the convolutional and ESM-2 heads of GSPA-Seq, the latter with embeddings interpolated from alignment hits (no ESM-2 forward pass was run for these proteins, and proteins without a hit receive no embedding channel), and writes one merged claim per protein–term pair; the released integrator (`gspa-2.1.0.jar integrate -batch-manifest`, six priors, Stage 2A relaxation) was run on these claims for 7,009 species-representative genomes and 4,592 further genomes, and rescued and suppressed protein–term pairs were counted by comparing the Stage 0 and final posteriors at 0.50 (Supplementary Section S7). Across whole-proteome benchmarks, donor-homology masking removes reference proteins sharing $\text{pident}(g, r) > \eta_{\text{id}}$ ($\eta_{\text{id}} \in \{100\%, 70\%, 50\%, 30\%\}$) or belonging to the test organism's genus or family across DIAMOND databases, ESM-2 and ProtT5 neighbor indices, FoldSeek structural templates, and calibration sets. In leave-reaction-out benchmarks, Track A masks sequence alignments and embedding donors with cosine similarity > 0.50 without reaction centroids, while Track B masks sequence alignments and retains Swiss-Prot ESM-2 reaction centroids.

We report official CAFA information-accretion-weighted f_w ($WF_{\max}$), protein-centric unweighted $F_{\max}$, pair-level micro $F_{\max}$, and minimum semantic distance $S_{\min}$ [1, 34, 74] over threshold grids $\theta \in \{0.01, \ldots, 0.99\}$ ($\{0.10, \ldots, 0.90\}$ in the eukaryotic stress test), alongside Hit@k and mean reciprocal rank (MRR) for candidate ranking, 95% paired bootstrap confidence intervals over proteomes, and two-sided exact sign tests over genomes or exact binomial McNemar tests over paired cases (Supplementary Section S5). All CPU benchmarks ran on dual-socket AMD EPYC 7763 nodes (8 threads per genome), with PLM, 16-kernel DeepGOCNN, ESM-2 3B, external SOTA GO baseline retraining, executed on NVIDIA A100-SXM4-80GB, NVIDIA H100-80GB, and NVIDIA H200-140GB GPUs, and Stage 3 language-model inference on two NVIDIA RTX 4090 GPUs.

The Genome-Scale Annotation Evaluation Framework (GAEF) summarizes structural proteome invariants at $\theta_{\text{eval}} = 0.50$ by combining essential function completeness C_{ess} ($w_{\text{ess}} = 0.30$), functional pathway/complex/process coherence C_{func} ($w_{\text{func}} = 0.40$), taxonomic and anatomical consistency $C_{\text{cons}} = \max(0, 1 - 0.01V(\mathcal{P}))$ ($w_{\text{cons}} = 0.30$), and functional resolution $C_{\text{res}} = \min(1, \text{meanIC}/10)$ ($w_{\text{res}} = 0.0$):

$$\text{GAEF}(\mathcal{P}) = \frac{\sum_{d \in \{\text{ess,func,cons,res}\}: C_d \neq \text{null}} w_d \, C_d}{\sum_{d: C_d \neq \text{null}} w_d}, \quad (8)$$

which equals $0.30C_{\text{ess}} + 0.40C_{\text{func}} + 0.30C_{\text{cons}}$ when all three structural dimensions are active. Because GAEF measures the structural invariants targeted by Stages 1 and 2, we report it as an internal structural audit score alongside independent held-out functional and experimental phenotype evaluations.

Data availability

All result tables underlying the figures and tables, the precomputed cross-ontology constraint tables (Supplementary Table S7), the benchmark manifests with checksums, and the provenance map linking every manuscript value to its result file and pipeline are in the repository at https://github.com/bio-ontology-research-group/gspa (release v2.1.0, CC-BY-4.0). The large inputs that the repository excludes, namely the ModelSEED-derived matrices and archived METEOR code required by the Stage 2B refiner and the whole-genome CAFA caches (proteomes, truth files, operons, gaps, InterProScan outputs, ESM-2 3B embeddings, retrained-baseline predictions, and the GSPA-Seq checkpoint), are available as checksummed archives at https://bio2vec.net/data/gspa/.

Code availability

The GSPA v2.1.0 software suite (JVM integration and evaluation engine, the Stage 2B stoichiometric locus refiner as a Python sidecar, the Stage 3 agent and ontology guardrail), the regeneration package `paper-reproducibility/` with one pipeline per benchmark, the table and figure generators, and the number checker are available under the GPL-3.0-or-later license at https://github.com/bio-ontology-research-group/gspa (release v2.1.0; `gspa-2.1.0.jar` is attached to the release).

Acknowledgments

This work was supported by King Abdullah University of Science and Technology (KAUST) baseline funding to

R.H.

Author contributions

T.A.-S., M.K., and R.H. conceived the study and designed the methodology. T.A.-S. and R.H. implemented the software, cross-ontology bridges, stoichiometric refiner, and benchmark suites. All authors analyzed the results and wrote the manuscript.

Competing interests

The authors declare no competing interests.

References

[1] Predrag Radivojac, Wyatt T. Clark, Tal Ronnen Oron, Alexandra M. Schnoes, Tobias Wittkop, Artem Sokolov, Kiley Graim, Christopher Funk, Karin Verspoor, Asa Ben-Hur, Gaurav Pandey, Jeffrey M. Yunes, Ameet S. Talwalkar, Sanna Repo, Michael L. Souza, Damiano Piovesan, Rong Zhong, Haixuan Cui, Nan Zhou, Melanie Kirac, Slobodan Vucetic, Aakrosh Ratan, Burkhard Rost, Julian Gough, Matthew Rogers, Franck Rapaport, Janet M. Thornton, Maria J. Martin, Amos Bairoch, Sean D. Mooney, Constance J. Jeffery, Iddo Friedberg, and Steven E. Brenner. A large-scale evaluation of computational protein function prediction. *Nature Methods*, 10 (3):221–227, 2013. doi: 10.1038/nmeth.2340. URL https://doi.org/10.1038/nmeth.2340.

[2] Nan Zhou, Yuxiang Jiang, Timothy R. Bergquist, Angela J. Lee, Balint Z. Kacsoh, Alex W. Crocker, Kaitlyn A. Lewis, George Georghiou, Huy N. Nguyen, Predrag Radivojac, and Iddo Friedberg. The CAFA challenge reports improved protein function prediction and new functional annotations for hundreds of genes through experimental screens. *Genome Biology*, 20 (1):244, 2019. doi: 10.1186/s13059-019-1835-8. URL https://doi.org/10.1186/s13059-019-1835-8.

[3] Michael Ashburner, Catherine A. Ball, Judith A. Blake, David Botstein, Heather Butler, J. Michael Cherry, Allan P. Davis, Kara Dolinski, Selina S. Dwight, Janan T. Eppig, Midori A. Harris, David P. Hill, Laurie Issel-Tarver, Andrew Kasarskis, Suzanna Lewis, John C. Matese, Joel E. Richardson, Martin Ringwald, Gerald M. Rubin, and Gavin Sherlock. Gene Ontology: tool for the unification of biology. *Nature Genetics*, 25(1):25–29, 2000. doi: 10.1038/75556.

[4] The Gene Ontology Consortium. The Gene Ontology knowledgebase in 2023. *Genetics*, 224(1):iyad031, 2023. doi: 10.1093/genetics/iyad031.

[5] Andrei Osterman and Ross Overbeek. Missing genes in metabolic pathways: a comparative genomics approach. *Current Opinion in Chemical Biology*, 7(2):238–251, 2003. doi: 10.1016/S1367-5931(03)00027-9. URL https://doi.org/10.1016/S1367-5931(03)00027-9.

[6] Peter D. Karp, Daniel Weaver, and Mario Latendresse. How accurate is automated gap filling of metabolic models? *BMC Systems Biology*, 12(1):59, 2018. doi: 10.1186/s12918-018-0593-7. URL https://doi.org/10.1186/s12918-018-0593-7.

[7] Madalina Giurgiu, Julian Reinhard, Barbara Brauner, Irmtraud Dunger-Kaltenbach, Gisela Fobo, Goar Frishman, Corinna Montrone, and Andreas Ruepp. Corum: the comprehensive resource of mammalian protein complexes – 2019. *Nucleic Acids Research*, 47 (D1):D559–D563, October 2018. ISSN 1362-4962. doi: 10.1093/nar/gky973. URL http://dx.doi.org/10.1093/nar/gky973.

[8] Birgit H M Meldal, Livia Perfetto, Colin Combe, Tiago Lubiana, João Vitor Ferreira Cavalcante, Hema Bye-A-Jee, Andra Waagmeester, Noemi del Toro, Anjali Shrivastava, Elisabeth Barrera, Edith Wong, Bernhard Mlecnik, Gabriela Bindea, Kalpana Panneerselvam, Egon Willighagen, Juri Rappsilber, Pablo Porras, Henning Hermjakob, and Sandra Orchard. Complex portal 2022: new curation frontiers. *Nucleic Acids Research*, 50(D1):D578–D586, October 2021. ISSN 1362-4962. doi: 10.1093/nar/gkab991. URL http://dx.doi.org/10.1093/nar/gkab991.

[9] Jennifer I. Deegan née Clark, Emily C. Dimmer, and Christopher J. Mungall. Formalization of taxon-based constraints to detect inconsistencies in annotation and ontology development. *BMC Bioinformatics*, 11: 530, 2010. doi: 10.1186/1471-2105-11-530. URL https://doi.org/10.1186/1471-2105-11-530.

[10] Christopher J. Mungall, Carlo Torniai, George V. Gkoutos, Suzanna E. Lewis, and Melissa A. Haendel. Uberon, an integrative multi-species anatomy ontology. *Genome Biology*, 13(1):R5, 2012. doi: 10.1186/gb-2012-13-1-r5. URL https://doi.org/10.1186/gb-2012-13-1-r5.

[11] Shanika L. Amarasinghe, Shian Su, Xueyi Dong, Luke Zappia, Matthew E. Ritchie, and Quentin Gouil. Opportunities and challenges in long-read sequencing data analysis. *Genome Biology*, 21(1):30, 2020. doi: 10.1186/s13059-020-1935-5.

[12] Glennis A. Logsdon, Mitchell R. Vollger, and Evan E. Eichler. Long-read human genome sequencing and its applications. *Nature Reviews Genetics*, 21(10): 597–614, 2020. doi: 10.1038/s41576-020-0236-x.

[13] Mikhail Kolmogorov, Derek M. Bickhart, Bahar Behsaz, Alexey Gurevich, Mikhail Rayko, Sung Bong Shin, Kristen Kuhn, Jeffrey Yuan, Evgeny Polevikov, Timothy P. L. Smith, and Pavel A. Pevzner. metaFlye: scalable long-read metagenome assembly using repeat graphs. *Nature Methods*, 17(11):1103–1110, 2020. doi: 10.1038/s41592-020-00971-x.

[14] Benjamin Buchfink, Klaus Reuter, and Hajk-Georg

Drost. Sensitive protein alignments at tree-of-life scale using diamond. *Nature Methods*, 18(4):366–368, 2021. doi: 10.1038/s41592-021-01101-x.

[15] Wenjun Li, Kathleen R O'Neill, Daniel H Haft, Michael DiCuccio, Vyacheslav Chetvernin, Azat Badretdin, George Coulouris, Farideh Chitsaz, Myra K Derbyshire, A Scott Durkin, Noreen R Gonzales, Marc Gwadz, Christopher J Lanczycki, James S Song, Narmada Thanki, Jiyao Wang, Roxanne A Yamashita, Mingzhang Yang, Chanjuan Zheng, Aron Marchler-Bauer, and Françoise Thibaud-Nissen. Refseq: expanding the prokaryotic genome annotation pipeline reach with protein family model curation. *Nucleic Acids Research*, 49(D1):D1020–D1028, December 2020. ISSN 1362-4962. doi: 10.1093/nar/gkaa1105. URL http://dx.doi.org/10.1093/nar/gkaa1105.

[16] Oliver Schwengers, Lukas Jelonek, Marius Alfred Dieckmann, Sebastian Beyvers, Jochen Blom, and Alexander Goesmann. Bakta: rapid and standardized annotation of bacterial genomes via alignment-free sequence identification: Find out more about bakta, the motivation, challenges and applications, here. *Microbial Genomics*, 7(11), November 2021. ISSN 2057-5858. doi: 10.1099/mgen.0.000685. URL http://dx.doi.org/10.1099/mgen.0.000685.

[17] Sean R. Eddy. Accelerated profile HMM searches. *PLoS Computational Biology*, 7(10):e1002195, 2011. doi: 10.1371/journal.pcbi.1002195.

[18] Philip Jones, David Binns, Hsin-Yu Chang, Matthew Fraser, Weizhong Li, Craig McAnulla, Hamish McWilliam, John Maslen, Alex Mitchell, Gift Nuka, Sebastien Pesseat, Antony F. Quinn, Amaia Sangrador-Vegas, Maxim Scheremetjew, Siew-Yit Yong, Rodrigo Lopez, and Sarah Hunter. Interproscan 5: genome-scale protein function classification. *Bioinformatics*, 30(9):1236–1240, January 2014. ISSN 1367-4803. doi: 10.1093/bioinformatics/btu031. URL http://dx.doi.org/10.1093/bioinformatics/btu031.

[19] Maxat Kulmanov, Mohammed Asif Khan, and Robert Hoehndorf. DeepGO: predicting protein functions from sequence and interactions using a deep ontology-aware classifier. *Bioinformatics*, 34(4):660–668, 2018. doi: 10.1093/bioinformatics/btx624.

[20] Maxat Kulmanov and Robert Hoehndorf. DeepGOPlus: improved protein function prediction from sequence. *Bioinformatics*, 36(2):422–429, 2020. doi: 10.1093/bioinformatics/btz595.

[21] Ronghui You, Shuwei Yao, Hiroshi Mamitsuka, and Shanfeng Zhu. DeepGraphGO: graph neural network for large-scale, multispecies protein function prediction. *Bioinformatics*, 37(Supplement_1):i262–i271, 2021. doi: 10.1093/bioinformatics/btab270.

[22] Yue Cao and Yang Shen. TALE: Transformer-based protein function annotation with joint sequence–label embedding. *Bioinformatics*, 37(18):2825–2833, 2021. doi: 10.1093/bioinformatics/btab178. URL https://doi.org/10.1093/bioinformatics/btab178.

[23] Zeming Lin, Halil Akin, Roshan Rao, Brian Hie, Zhongkai Zhu, Wenting Lu, Nikita Smetanin, Robert Verkuil, Ori Kabeli, Yaniv Shmueli, et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637): 1123–1130, 2023. doi: 10.1126/science.ade2574.

[24] Ahmed Elnaggar, Michael Heinzinger, Christian Dallago, Ghalia Rehawi, Yu Wang, Llion Jones, Tom Gibbs, Tamas Feher, Christoph Angerer, Martin Steinegger, Debsindhu Bhowmik, and Burkhard Rost. ProtTrans: Toward understanding the language of life through self-supervised learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(10): 7112–7127, 2022. doi: 10.1109/TPAMI.2021.3095381.

[25] Qianmu Yuan, Junjie Xie, Jiancong Xie, Huiying Zhao, and Yuedong Yang. Fast and accurate protein function prediction from sequence through pretrained language model and homology-based label diffusion. *Briefings in Bioinformatics*, 24(3):bbad117, 2023. doi: 10.1093/bib/bbad117. URL https://doi.org/10.1093/bib/bbad117.

[26] Maxat Kulmanov, Francisco J. Guzmán-Vega, Paula Duek Roggli, Lydie Lane, Stefan T. Arold, and Robert Hoehndorf. Protein function prediction as approximate semantic entailment. *Nature Machine Intelligence*, 6(2):220–228, 2024. doi: 10.1038/s42256-024-00795-w. URL https://doi.org/10.1038/s42256-024-00795-w.

[27] Tianhao Yu, Haiyang Cui, Jianan Li, Yang Yang, Hui Xiong, Zixuan Cang, and Huimin Zhao. Enzyme function prediction using contrastive learning. *Science*, 379(6639):1358–1363, 2023. doi: 10.1126/science.adf2470. URL https://doi.org/10.1126/science.adf2470.

[28] Vladimir Gligorijević, P. Douglas Renfrew, Tomasz Kosciolek, Julia Koehler Leman, Daniel Berenberg, Tommi Vatanen, Chris Chandler, Ian Mavric, Thomas Hill, Kyunghyun Cho, and Richard Bonneau. Structure-based protein function prediction using graph convolutional networks. *Nature Communications*, 12:3168, 2021. doi: 10.1038/s41467-021-23303-9. URL https://doi.org/10.1038/s41467-021-23303-9.

[29] Michel van Kempen, Stephanie S. Kim, Charlotte Tumescheit, Milot Mirdita, Jeongjae Lee, Cameron L. M. Gilchrist, Johannes Söding, and Martin Steinegger. Fast and accurate protein structure search with Foldseek. *Nature Biotechnology*, 42(2):243–246, 2024. doi: 10.1038/s41587-023-01773-0.

[30] Yuting Ding, Jian Wang, Meng Li, Jiawei Peng, Shuai Li, Jijun Tang, and Fei Guo. Accurate protein function prediction with domain-guided structure information. *Nature Communications*, 16(1): 70, 2025. doi: 10.1038/s41467-024-54816-8. URL https://doi.org/10.1038/s41467-024-54816-8.

[31] Ronghui You, Zihan Zhang, Yi Xiong, Fengzhu Sun, Hiroshi Mamitsuka, and Shanfeng Zhu. GOLabeler: improving sequence-based large-scale protein function prediction by learning to rank. *Bioinformatics*, 34(14): 2465–2473, 2018. doi: 10.1093/bioinformatics/bty130.

[32] Ronghui You, Shuwei Yao, Yi Xiong, Zhi-Ping Huang, Fengzhu Sun, Hiroshi Mamitsuka, and Shanfeng Zhu. NetGO: improving large-scale protein function prediction with graph-preserving learning and network integration. *Nucleic Acids Research*, 47(W1): W397–W402, 2019. doi: 10.1093/nar/gkz388. URL https://doi.org/10.1093/nar/gkz388.

[33] Shaojun Wang, Ronghui You, Yunjia Liu, Yi Xiong, and Shanfeng Zhu. NetGO 3.0: Protein language model improves large-scale functional annotations. *Genomics, Proteomics & Bioinformatics*, 21(2):349–358, 2023. doi: 10.1016/j.gpb.2023.04.001. URL https://doi.org/10.1016/j.gpb.2023.04.001.

[34] Yuxiang Jiang, Tal Ronnen Oron, Wyatt T. Clark, Atreya R. Bankapur, Daniel D'Andrea, Rosalba Lepore, Christopher S. Funk, Indika Kahanda, Karin M. Verspoor, Asa Ben-Hur, David Chi-Eng Koo, Dan Penfold-Brown, Dennis Shasha, Noah M. Young, Richard Bonneau, Björn Wallner, Nereo Atropo, Florence Martin, Antonio Puertas-Gallardo, Jose M. Carazo, Pedro Larrañaga, Concha Bielza, Luigi Cerulo, Alexandre Chigaev, Bruce S. Edwards, Larry A. Sklar, Emmanuel Charbonnier, François Enault, Angela Del Pozo, José M. Fernández, Michael L. Tress, Amar Bhaskar, Arne Elofsson, Shaojun Wang, Shanfeng Zhu, Ronghui You, Zhi-Ping Huang, Shanshan Dai, Shuwei Yao, Weidong Tian, S. Radha, Aakrosh Ratan, Burkhard Rost, Julian Gough, Predrag Radivojac, and Iddo Friedberg. An expanded evaluation of protein function prediction methods shows an improvement in accuracy. *Genome Biology*, 17(1):184, 2016. doi: 10.1186/s13059-016-1037-6. URL https://doi.org/10.1186/s13059-016-1037-6.

[35] Thomas Dandekar, Berend Snel, Martijn Huynen, and Peer Bork. Conservation of gene order: a fingerprint of proteins that physically interact. *Trends in Genetics*, 14(9):334–337, 1998. doi: 10.1016/S0168-9525(98)01558-3. URL https://doi.org/10.1016/S0168-9525(98)01558-3.

[36] Ross Overbeek, Michael Fonstein, Mark D'Souza, Gordon D. Pusch, and Natalia Maltsev. The use of gene clusters to infer functional coupling. *Proceedings of the National Academy of Sciences of the United States of America*, 96(6):2896–2901, 1999. doi: 10.1073/pnas.96.6.2896. URL https://doi.org/10.1073/pnas.96.6.2896.

[37] Matteo Pellegrini, Edward M. Marcotte, Michael J. Thompson, David Eisenberg, and Todd O. Yeates. Assigning protein functions by comparative genome analysis: Protein phylogenetic profiles. *Proceedings of the National Academy of Sciences of the United States of America*, 96(8):4285–4288, 1999. doi: 10.1073/pnas.96.8.4285. URL https://doi.org/10.1073/pnas.96.8.4285.

[38] Morgan N. Price, Katherine H. Huang, Eric J. Alm, and Adam P. Arkin. A novel method for accurate operon predictions in all sequenced prokaryotes. *Nucleic Acids Research*, 33(3):880–892, 2005. doi: 10.1093/nar/gki232. URL https://doi.org/10.1093/nar/gki232.

[39] Yunha Hwang, Andre L. Cornman, Elizabeth H. Kellogg, Sergey Ovchinnikov, and Peter R. Girguis. Genomic language model predicts protein co-regulation and function. *Nature Communications*, 15: 2888, 2024. doi: 10.1038/s41467-024-46947-9. URL https://doi.org/10.1038/s41467-024-46947-9.

[40] Andre Cornman, Jacob West-Roberts, Antonio Pedro Camargo, Simon Roux, Martin Beracochea, Milot Mirdita, Sergey Ovchinnikov, and Yunha Hwang. The OMG dataset: An open metagenomic corpus for mixed-modality genomic language modeling. *bioRxiv*, 2024. doi: 10.1101/2024.08.14.607850. URL https://doi.org/10.1101/2024.08.14.607850.

[41] Daulet Toibazar, Maxat Kulmanov, and Robert Hoehndorf. Context-based protein function prediction in bacterial genomes. *bioRxiv*, 2024. doi: 10.1101/2024.10.14.618363. URL https://doi.org/10.1101/2024.10.14.618363.

[42] Maciej Wiatrak, Ramon Viñas Torné, Maria Ntemourtsidou, Adam Dinan, David C. Abelson, Divya Arora, Maria Brbić, Aaron Weimann, and R. Andres Floto. A contextualised protein language model reveals the functional syntax of bacterial evolution. *bioRxiv*, 2025. doi: 10.1101/2025.07.20.665723. URL https://doi.org/10.1101/2025.07.20.665723.

[43] Michelle L. Green and Peter D. Karp. A Bayesian method for identifying missing enzymes in predicted metabolic pathway databases. *BMC Bioinformatics*, 5:76, 2004. doi: 10.1186/1471-2105-5-76. URL https://doi.org/10.1186/1471-2105-5-76.

[44] Samuel M. D. Seaver, Filipe Liu, Qizhi Zhang, Jose P. Roitberg, José P. Faria, and Christopher S. Henry. The ModelSEED biochemistry database for the integration of metabolic annotations and the reconstruction, comparison and analysis of metabolic models for plants, fungi and microbes. *Nucleic Acids Research*, 49(D1):D575–D588, 2021. doi: 10.1093/nar/gkaa746. URL https://doi.org/10.1093/nar/gkaa746.

[45] Daniel Machado, Sergej Andrejev, Melanie Tramontano, and Kiran Raosaheb Patil. Fast automated reconstruction of genome-scale metabolic models for microbial species and communities. *Nucleic Acids Research*, 46(15):7542–7553, 2018. doi: 10.1093/nar/gky537. URL https://doi.org/10.1093/nar/gky537.

[46] Johannes Zimmermann, Christoph Kaleta, and Silvio

Waschina. gapseq: informed prediction of bacterial metabolic pathways and reconstruction of accurate metabolic models. *Genome Biology*, 22(1):81, 2021. doi: 10.1186/s13059-021-02295-1. URL https://doi.org/10.1186/s13059-021-02295-1.

[47] Morgan N. Price, Adam M. Deutschbauer, and Adam P. Arkin. GapMind: Automated annotation of amino acid biosynthesis. *mSystems*, 5(3):e00291–20, 2020. doi: 10.1128/mSystems.00291-20.

[48] Kexin Niu, Maxat Kulmanov, and Robert Hoehndorf. METEOR: joint genome-scale reconstruction and enzyme prediction. In *Pacific Symposium on Biocomputing 2027 (Session: Biological Molecular Function: Beyond Homology; preprint: bioRxiv 2026.02.01.701942)*, volume 32. World Scientific, 2027. doi: 10.64898/2026.02.01.701942.

[49] Christopher J. Mungall, Michael Bada, Tanya Z. Berardini, Jennifer Deegan, Amelia Ireland, Midori A. Harris, David P. Hill, and Jane Lomax. Cross-product extensions of the Gene Ontology. *Journal of Biomedical Informatics*, 44(1):80–86, 2011. doi: 10.1016/j.jbi.2010.02.002. URL https://doi.org/10.1016/j.jbi.2010.02.002.

[50] Laurel Cooper, Ramona L. Walls, Justin Elser, Maria A. Gandolfo, Dennis W. Stevenson, Barry Smith, Justin Preece, and Pankaj Jaiswal. The Plant Ontology as a tool for comparative plant anatomy and development. *International Journal of Plant Sciences*, 174(1):41–53, 2013. doi: 10.1086/668614.

[51] Alexander D. Diehl, Terrence F. Meehan, Yvonne M. Bradford, Matthew H. Brush, Wasila M. Dahdul, David S. Dougall, Yongqun He, David Osumi-Sutherland, Alan Ruttenberg, Sirarat Sarntivijai, Ceri E. Van Slyke, Nicole A. Vasilevsky, Melissa A. Haendel, Judith A. Blake, and Christopher J. Mungall. The Cell Ontology 2016: enhanced content, modularization, and ontology interoperability. *Journal of Biomedical Semantics*, 7:44, 2016. doi: 10.1186/s13326-016-0088-0.

[52] Zhizheng Wang, Qiao Jin, Chih-Hsuan Wei, Shubo Tian, Po-Ting Lai, Qingqing Zhu, Chi-Ping Day, Christina Ross, Robert Leaman, and Zhiyong Lu. Geneagent: self-verification language agent for gene-set analysis using domain databases. *Nature Methods*, 22 (3):489–499, 2025. doi: 10.1038/s41592-025-02748-6. Preprint: arXiv:2405.16205.

[53] J. Harry Caufield, Harshad Hegde, Vincent Emonet, Nomi L. Harris, Marcin P. Joachimiak, Nicolas Matentzoglu, HyeongSik Kim, Sierra Moxon, Justin T. Reese, Melissa A. Haendel, Peter N. Robinson, and Christopher J. Mungall. Structured prompt interrogation and recursive extraction of semantics (spires): a method for populating knowledge bases using zero-shot learning. *Bioinformatics*, 40(3):btae104, 2024. doi: 10.1093/bioinformatics/btae104. Preprint: arXiv:2304.02711.

[54] Nathan E Lewis, Kim K Hixson, Tom M Conrad, Joshua A Lerman, Pep Charusanti, Ashoka D Polpitiya, Joshua N Adkins, Gunnar Schramm, Samuel O Purvine, Daniel Lopez-Ferrer, et al. Omic data from evolved *E. coli* are consistent with computed optimal growth from genome-scale models. *Molecular Systems Biology*, 6(1):390, 2010. doi: 10.1038/msb.2010.47.

[55] Valentyn Bezshapkin, Filip Schymik, Piotr Kucharski, Paweł Szczerbiak, Jakub W. Wojciechowski, Łukasz M. Szydłowski, and Tomasz Kościołek. Scaling and democratising structure-based protein function prediction with metagenomic-deepFRI. *bioRxiv*, page 2026.04.27.720983, 2026. doi: 10.64898/2026.04.27.720983. URL https://doi.org/10.64898/2026.04.27.720983.

[56] Nicolas Buton, François Coste, and Yann Le Cunff. Predicting enzymatic function of protein sequences with attention. *Bioinformatics*, 39(10):btad620, 2023. doi: 10.1093/bioinformatics/btad620.

[57] Gi Bae Kim, Ji Yeon Kim, Jong An Lee, Charles J Norsigian, Bernhard O Palsson, and Sang Yup Lee. Functional annotation of enzyme-encoding genes using deep learning with transformer layers. *Nature Communications*, 14(1):7370, 2023. doi: 10.1038/s41467-023-43216-z.

[58] Rund Tawfiq, Maxat Kulmanov, and Robert Hoehndorf. Beyond protein functions: evaluating completeness, coherence, and consistency of genome-scale function annotations. *Briefings in Bioinformatics*, 2026. doi: 10.1093/bib/bbag336.

[59] Morgan N Price, Kelly M Wetmore, R Jordan Waters, Mark Callaghan, Jayashree Ray, Hualan Liu, Jennifer V Kuehl, Ryan A Melnyk, Jacob S Lamson, Yumi Suh, et al. Mutant phenotypes for thousands of bacterial genes of unknown function. *Nature*, 557(7706): 503–509, 2018. doi: 10.1038/s41586-018-0124-0.

[60] Lorenz Christian Reimer, Joaquim Sardà Carbasse, Julia Koblitz, Christian Ebeling, Adam Podstawka, and Jörg Overmann. BacDive in 2022: the knowledge base for standardized bacterial and archaeal data. *Nucleic Acids Research*, 50(D1):D741–D746, 2022. doi: 10.1093/nar/gkab961.

[61] Isabel Schober, Julia Koblitz, Joaquim Sardà Carbasse, Christian Ebeling, Marvin Leon Schmidt, Adam Podstawka, Rohit Gupta, Vinodh Ilangovan, Javad Chamanara, Jörg Overmann, and Lorenz Christian Reimer. BacDive in 2025: the core database for prokaryotic strain data. *Nucleic Acids Research*, 53 (D1):D748–D756, 2025. doi: 10.1093/nar/gkae959.

[62] Martin Hunt, Marcelo D. T. Torres, Nabil-Fareed Alikhan, Daniel Anderson, Maria Luiza Andreani, Josefin Blom, George Bouras, Fiona S. L. Brinkman, Laura M. Carroll, Matthew A. Croxen, R. Andres Floto, Michael B. Hall, Jane Hawkey, Samuel T. Horsfield, Baofeng Jia, Jake A. Lacey, Hyun-Su Lee, Leandro Lima, Neil MacAlasdair, Su-

daraka Mallawaarachchi, William Matlock, Ahmed M. Moustafa, Robert Petit, Vignesh Ramnath, Vishnu Raghuram, Matthew J. Russell, Theo Sanderson, Timo Saratto, Oliver Schwengers, Torsten Seemann, Liam P. Shaw, Wei Shen, Nicholas Thomson, Gerry Tonkin-Hill, Jackie Toussaint, Thanh Le Viet, Johanna von Wachsmann, Fangping Wan, Aaron Weimann, Rachel M. Wheatley, Maciej Wiatrak, Ouli Xie, Cesar de la Fuente-Nunez, John A. Lees, and Zamin Iqbal. AllTheBacteria: a community resource empowers biology and discovers novel peptide antibiotics. *bioRxiv*, page 2024.03.08.584059, 2024. doi: 10.1101/2024.03.08.584059. URL https://doi.org/10.1101/2024.03.08.584059.

[63] Dingyi Rong, Bozitao Zhong, Wenzhuo Zheng, Liang Hong, and Ning Liu. Autoregressive enzyme function prediction with multi-scale multi-modality fusion. *Briefings in Bioinformatics*, 26(5):bbaf476, 2025. doi: 10.1093/bib/bbaf476.

[64] Yidong Song, Qianmu Yuan, Sheng Chen, Yuansong Zeng, Huiying Zhao, and Yuedong Yang. Accurately predicting enzyme functions through geometric graph learning on ESMFold-predicted structures. *Nature Communications*, 15(1):8180, 2024. doi: 10.1038/s41467-024-52533-w.

[65] Karel van der Weg, Erinc Merdivan, Marie Piraud, and Holger Gohlke. TopEC: prediction of Enzyme Commission classes by 3D graph neural networks and localized 3D protein descriptors. *Nature Communications*, 16(1):2737, 2025. doi: 10.1038/s41467-025-57324-5.

[66] Theo Sanderson, Maxwell L. Bileschi, David Belanger, and Lucy J. Colwell. ProteInfer, deep networks for protein functional inference. *eLife*, 12:e80942, 2023. doi: 10.7554/eLife.80942.

[67] Maxat Kulmanov and Robert Hoehndorf. DeepGOWeb: fast and accurate protein function prediction on the (Semantic) Web. *Nucleic Acids Research*, 49 (W1):W134–W140, 2021. doi: 10.1093/nar/gkab373. URL https://doi.org/10.1093/nar/gkab373.

[68] Michael Heinzinger, Konstantin Weissenow, Joaquin Gomez Sanchez, Adrian Henkel, Milot Mirdita, Martin Steinegger, and Burkhard Rost. Bilingual language model for protein sequence and structure. *NAR Genomics and Bioinformatics*, 6(4):lqae150, 2024. doi: 10.1093/nargab/lqae150.

[69] Aaron Weimann, Kyra Mooren, Jochen Frank, Phillip B. Pope, Andreas Bremges, and Alice C. McHardy. From genomes to phenotypes: Traitar, the microbial trait analyzer. *mSystems*, 1(6):e00101–16, 2016. doi: 10.1128/mSystems.00101-16.

[70] Julia Koblitz, Lorenz Christian Reimer, Rüdiger Pukall, and Jörg Overmann. Predicting bacterial phenotypic traits through improved machine learning using high-quality, curated datasets. *Communications Biology*, 8(1):897, 2025. doi: 10.1038/s42003-025-08313-3.

[71] Maria Littmann, Michael Heinzinger, Christian Dallago, Tobias Olenyi, and Burkhard Rost. Embeddings from deep learning transfer GO annotations beyond homology. *Scientific Reports*, 11(1):1160, 2021. doi: 10.1038/s41598-020-80786-0. URL https://doi.org/10.1038/s41598-020-80786-0.

[72] Eleonora Giunchiglia and Thomas Lukasiewicz. Coherent hierarchical multi-label classification networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 9662–9673, 2020. doi: 10.48550/arXiv.2010.10151. URL https://doi.org/10.48550/arXiv.2010.10151.

[73] Martin Steinegger and Johannes Söding. Clustering huge protein sequence sets in linear time. *Nature Communications*, 9(1):2542, 2018. doi: 10.1038/s41467-018-04964-5. URL https://doi.org/10.1038/s41467-018-04964-5.

[74] Wyatt T. Clark and Predrag Radivojac. Information-theoretic evaluation of predicted ontological annotations. *Bioinformatics*, 29(13):i53–i61, 2013. doi: 10.1093/bioinformatics/btt228. URL https://doi.org/10.1093/bioinformatics/btt228.